

**BRAILLE CONNECT: A MOBILE APPLICATION FOR VISUALLY IMPAIRED  
STUDENTS**



A Thesis/Project Study

**Presented to the Faculty of BTVTED Program**

**City College of Davao**

Km. 10 Catalunan Pequeño, Davao City

In Partial Fulfillment of the Requirements for the Degree of

**Bachelor of Technical-Vocational Teacher Education**

**Major in Computer Programming**

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## ABSTRACT

This study aimed to design, develop, and evaluate Braille Connect, a mobile application designed to support and assist the visually impaired students in inclusive learning. It was directed by Assistive Technology Theory and the principles of Inclusive Education, which led to the creation of the study aimed at improving learning, accessibility, and independence of the users within the classroom environment. The application includes word-to-braille and braille-to-word translation features, image scanning and object and color recognition, as well as audio feedback, such as the TalkBack feature and uploading files. Using developmental research design, the system was evaluated by twenty (20) respondents composed of blind students, teachers, and parents. Results showed high mean scores in functionality (4.78), usability (4.65), and effectiveness (4.71). Such findings demonstrate that the application is efficient, easy to use and navigate in helping to learn and be environmentally conscious. These findings conclude that Braille Connect Mobile Application is a reliable assistive tool that promotes accessibility, inclusive learning, and student independence.

**Keywords:** *Text-to-Speech, Braille, Translation, Mobile Application, Audio, Blind, Visually Impaired, Scan, Image, Software, Capture, Computer Programming*

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**APPROVAL SHEET**

This technology research titled, “**Braille Connect: A Mobile Application for Visually Impaired Students**,” of the City College of Davao, prepared and submitted by **Neil Francis T. Arnaiz and Zaldy A. Dagohoy** in partial fulfillment of the requirements of the degree of Bachelor of Technical Vocational Teacher Education major in Computer Programming, has been examined and is recommended for approval and acceptance.

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**APPROVED** by the technology research committee on oral examination with a grade of **PASSED**.

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N.F.T.A.  
Z.A.D.

## DEDICATION

*The researchers dedicate this study to their families and friends, who have served as backbones and foundations all throughout the research process.*

*To our respondents who willingly offered their time, experiences, and insights for this research. We dedicate this to all of you for your perseverance and positive outlook on life despite the challenges you have been through.*

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N.F.T.A.  
Z.A.D.

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## **Chapter I**

### **INTRODUCTION**

#### **Background of the Study**

Although inclusive education has been introduced with great effort in the public primary and secondary schools in Ethiopia, the field remains fraught with a lot of challenges, especially for students with visual impairment (Belay & Yihun, 2020). The study reveals that, despite the schools making efforts to accommodate inclusive education, the level of material and qualified personnel to meet the schools' needs is still not adequate.

Challenges for visually impaired students are not only present in basic education but also in higher education. In Malaysia, Amin et al., (2021) interviewed five students about their experiences in the institution. The results showed that students with visual impairments face five major challenges, including issues with accessibility and difficulties in real-time learning. Despite the extra attention and effort provided, educational experts still struggle to make them learn together with normal students. Thus, accessibility for visually impaired students remains a challenge in many countries.

Even when The United Nations Convention on the Rights of Persons with Disabilities (CRPD) guarantees the disabled access to education, which includes the provision of braille teaching to blind persons. However, Harris et al., (2022) suggest that there is evidence that braille teaching was discontinued due to the opinion that it is not an effective learning medium for blind pupils. This implies

that visually impaired students that are using braille still show ineffectiveness in learning.

Moreover, The Higher Education Institution in Cebu, Philippines conducted a study about the challenges and opportunities of visually impaired students in a public institution. Findings show that students with visual impairment experienced opportunities only in accessibility, therefore it is recommended that schools should enhance their services and resources, most especially when giving tasks and activities for blind individuals (Bacalla et al., 2024).

Prior to the pandemic, the Commission on Higher Education (2017) provided higher education students with supporting services and equipment such as the provision of a companion, a qualified reader, and a shadow teacher (also called an assistant) for additional services to meet their special needs, academic concerns, and the use of assistive equipment such as Braille (Ocampo et al., 2023). However, it is still not guaranteed that the learners will keep up with normal students in terms of learning and studying. Moreover, according to the UN (2020), children with visual impairments are considered vulnerable learners who should get equity, inclusion, and support. Nevertheless, they are frequently left behind due to lack of digital skills and have limited access to technology and connectivity (Ocampo et al., 2023).

A research study conducted by Ignacio (2015) in Saint Louis University in Baguio City, Philippines about the lived experiences of visually impaired tertiary students. The result shows both positive and negative experiences faced by the

students. Due to their negative experience, the blind students are not always comfortable in their classes. At times they feel discriminated against, embarrassed, and worried due to some undesirable practices, activities, or events in their classes.

However, visually impaired students are not the only ones struggling in the field of education, as well as the teachers. A study was conducted in Davao Oriental, Philippines, where fifteen teachers were selected to participate. The findings revealed that despite the teachers' strong commitment to inclusive education, many of them are constrained by a variety of factors, including insufficient training, lack of AT, and inflexibility in assessment processes (Lazaga, 2025). This indicates that a greater support system is needed, more training, and resources that are based on real-world scenarios.

Meanwhile, a study in Davao City found out that some teachers report gaps in training, limited resources, and inconsistent application of adaptive strategies for learners with disabilities, issues that directly affect visually impaired students' schooling (Cagas et al., 2023). Teachers are also adjusting to the education of visually impaired students, making them experience more loaded schoolwork.

Furthermore, the study of Villarente and Judilla (2025) in the University of Southeastern Philippines, Davao City, indicated that although inclusive education has developed and progressed all over the world and in the Philippines, there is a drop in the literacy rate of braille. The trend is blamed on the over-dependence on the use of audio-based technologies such as screen reading and audiobooks

and the resulting downfall of using the traditional Braille instruction itself. This implies that utilizing Braille script requires proper knowledge and skills. Traditional Braille instruction needs to be developed in order to keep up with the advancements.

Therefore, despite various research and studies highlighting the challenges and opportunities of visually impaired students in inclusive education, still several gaps are evident. Past research shows that visually impaired students in higher education are given less attention than those in basic and secondary education. Existing studies also emphasize student challenges but provide insufficient examination of teachers' preparedness and adaptive strategies in handling visually impaired learners. Furthermore, the decline in Braille literacy and the growing reliance on audio-based technologies have been documented; however, still some research is not enough to address the issues. Finally, most research is largely concentrated in urban institutions, creating a gap in understanding regional disparities in the implementation of inclusive education.

## **Objectives of the Study**

This project study sought to accomplish the following:

1. Design and Develop Braille Connect: A Mobile Application for Visually Impaired Students
2. Test the functionality of the project in terms of:
  - 2.1 Accuracy of word-to-braille and braille-to-word translation
  - 2.2 Accuracy of object recognition from scanned images
  - 2.3 Clarity and audibility of the talkback voice output
  - 2.4 Clarity and understandability of generated image description
  - 2.5 Ability of the Braille Connect Application to upload file formats such as .pdf, .docs, .docx, and .txt files
3. Assess the Usability of the project in terms of:
  - 3.1 Accessibility of the interface for visually impaired users
  - 3.2 Ease of understanding and operating the application controls
  - 3.3 Ease of using the scanning feature for blind students
  - 3.4 Clarity and ease of following the audio output
  - 3.5 System stability during extended use (absence of lag, crashes, or performance issues)

4. Evaluate the Effectiveness of the project in terms of:

- 4.1 Helping visually impaired users identify in their surroundings
- 4.2 Supporting visually impaired students in their learning activities
- 4.3 Assisting teachers and parents in guiding visually impaired students
- 4.4 Improving users' awareness of their environment
- 4.5 Promoting independence among visually impaired students

### **Significance of the Study**

This project, "Braille Connect: A Mobile Application for Visually Impaired Students," will serve as an insight and reference for the following:

#### **Educational Institutions**

The researchers intend to utilize innovative technology to provide and support colleges, universities, and other learning institutions to promote inclusive education.

**Visually Impaired Students**

This project aims to allow visually impaired students by giving them an efficient tool for learning and communication.

**Teachers**

The researchers desire to support teachers in delivering lessons to visually impaired learners through accessible technology.

**Parent**

This project aims to support parents of visually challenged students by offering them a trustworthy educational aid that enhances their children's learning experience at home.

**Future Researcher**

This project offers a foundation for future researchers who wish to explore advancements in assistive technologies

## **Scope and Delimitation**

This research is devoted to the design and development of Braille Connect, a mobile application for visually impaired students. The application will be used to take pictures, identify items, create audio tags, and perform word-to-Braille and Braille-to-word translations. It is meant to assist visually impaired students in their day-to-day lives and education by improving accessibility, environmental consciousness, and self-sufficiency. The application is also tested on the basis of functionality, usability, and effectiveness, where visually impaired students, teachers, and family respondents are used. Moreover, the user's age range is 12 and above because during this age the users already know how to operate a mobile phone.

Nevertheless, this study is subject to certain delimitation. The application only supports image recognition of objects, basic text translation, and audio output. It is not compatible with further features like video processing, real-time navigation, or the interpretation of the complex mathematical and graphical content. The application can be affected differently by environmental conditions and device specifications, which can affect its performance. Also, the application is only available to Android devices and could not be installed yet to iOS devices. Moreover, the assessment is limited to a small sample of respondents in the chosen learning institutions. Such constraints guarantee a narrow and realistic development and evaluation.

## **Chapter II**

### **CONCEPTUAL FRAMEWORK**

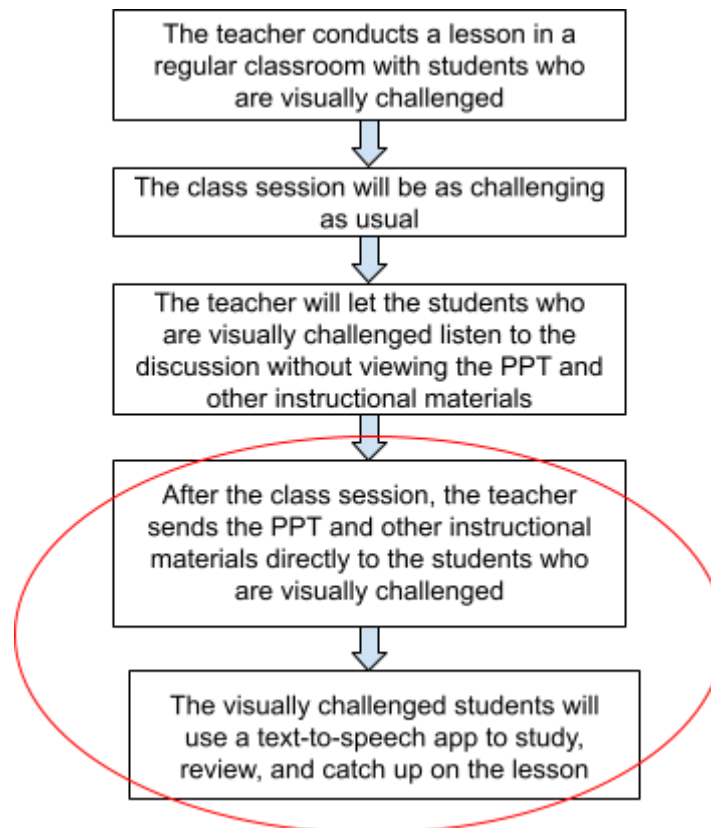
#### **Analysis of the Problem**

The diagram describes the traditional method and the existing practice in teaching visually impaired students in an inclusive education setting.

As mentioned in the previous chapter, Belay and Yihun (2020) and Amin et al. (2021) emphasizes that despite the efforts for strengthening inclusive education, visually impaired students continue to face barriers in accessibility, resources, and support systems. Studies reveal that there are many schools that still lack assistive devices and qualified professionals, making it difficult for blind learners to cope with academic demands. Due to over-reliance on audio-based technologies, braille literacy is declining, which reduces students' ability to develop independent reading and writing skills (Harris et al., 2022; Villarente & Judilla, 2025).

Additionally, Lazaga (2025) and Cagas et al. (2023) suggest that limited training opportunities, inadequate adaptive tools, and heavier workloads when handling learners with disabilities are some of the many challenges that teachers encounter as well. Psychosocial issues such as discrimination and exclusion further affect students' participation and well-being (Ignacio, 2015). Moreover, there are still gaps in understanding higher education contexts and regional

disparities since most research often focuses on urban institutions and basic education (Bacalla et al., 2024). Consequently, these factors mark the gap between actual practices and policy aspirations in applying inclusive education for the visually impaired.



*Figure 1. Process Flow of the Existing Process*

The figure above shows the practices that explain why the existing pedagogical method might not fully cater the visually challenged students inclusively.

In the first and second steps, the teacher conducts a lesson in a regular classroom with students who have visual impairment, which can make the class session much more challenging for teachers.

In the third step, most of the teachers simply allow students who have visual impairments to listen to the discussion, which places them in a much more challenging situation, limiting opportunities for active participation.

In the fourth and the last step, the teacher sends the PPT and other instructional materials directly to the said students and allows the students to independently utilize a text-to-speech application to convert all instructional materials into auditory form.

## **Review of Related Literature**

The study of Kana and Hagos (2024) found out that a systematic review examined the barriers to learning braille and used it as a mode of instruction and assessing learners with visual impairments. It identifies several challenges that contribute to the spread and development of braille in schools, like reliance of visually impaired students on voice recognition, teachers' lack of knowledge in Braille, scarcity of Braille-printed materials, and educational leadership-related factors such as lack of policy and lack of Braille technologies.

Moreover, students with visual impairments face several issues and challenges due to lack of visual skills. This includes issues with reading and assessing instructional materials. Thereby, teachers of special education had to face various challenges as well while teaching and giving instructions to their students with vision loss (Munir et al., 2024). Braille Script serves as a

fundamental medium of communication for visually impaired people however, its interpretation by non-experts is a major challenge to education and day to day interaction, hindering inclusivity (S et al., 2025). In Sri Lanka, Vithanage (2021) reveals that students with visual impairments, who work with braille, suffer a lot, as well as their teachers and facilitators as they cannot understand braille language.

Aslan (2025) agreed that writing in braille and utilizing writing skills are challenging for visually impaired students. These students may face specific issues and make errors in their Braille writing. These errors include the inappropriate use of symbols as well as errors related to spacing. This implies that even students with better knowledge in braille reading and writing still face challenges and require action.

While various software and websites enable the translation of information into any language, providing flexibility and accessibility across diverse linguistic preferences, making people grow their knowledge about a certain context. However, the visually impaired have no standard means to translate information to braille, which has caused a significant information gap for them, making them behind on the new information due to the unavailability of the information in their language (Translation System for the Visually Impaired From English to Braille, 2024).

Although braille is widely used in many text formats, such as portable and refreshable devices, its applications in the classroom are limited. The context of

the classroom board is still not fully mapped for the visually impaired students, despite the fact that there have been many refreshable braille devices available in recent decades (Wearable Haptic Braille Device for Enhancing Classroom Learning, 2023).

Additionally, access to a Teacher of Students with Visual Impairments (TVI) is one of the biggest problems in learning braille. These are professionals who teach visually impaired students on how to use braille. Furthermore, issues include the difference between print and braille and the different views needed for a typically sighted TVI and students with visual impairments (World Braille Day: Research on Teaching Braille to Students With Visual Impairments | IES, 2021).

Citizen Specialty Hospital (n.d.) argued that there is an area that is especially pronounced in underfunded schools and developing countries, where the cost and time involved with production can make it less of a priority to create braille materials. Thus, textbooks and other educational resources are not translated into braille in a timely fashion. This puts visually impaired students far behind their peers and unable to get the same quality of instruction.

A study found that second-grade students had greater difficulties reading Braille than writing it, that higher education institutions lacked haptic training, and that there were insufficient training programs for teaching and studying Braille (Iqbal et al., 2022).

Consequently, braille literacy skills have decreased in special primary schools for students with visual impairments. Teachers believe that the following

factors contributed to this decline in Braille standards: the rise in the number of students with multiple disabilities, the student-teacher ratio, insufficient teacher preparation, and incompetent teachers. Additionally, the results showed that educators lacked specialized training in Early Childhood Education (ECD) for students with visual impairments (Munyi, 2017).

Ampweya (2022) evaluated learners with visual impairments couldn't be able to take the subjects like Mathematics, Accounting, Geography, Physics as teachers were not properly trained to teach the respective subjects in braille to visually impaired learners. Also, visually impaired students are not able to use computers during ICT classes as there is no any program that caters and supports the teaching and learning of visually impaired students .

Although the preservation and maintenance of braille literature and documents are very important. However, hard copy braille is heavy, bulky, and requires a lot of space to store and maintain. Also, the materials that are available only in hard copy bear a threat of misplacing, or worse, losing them (Optical Braille Recognition and Its Correspondence in the Conversion of Braille Script to Text — a Literature Review, 2016). Blind and Visually Impaired people face various challenges in writing as they typically use traditional tools such as Slate, Stylus and Braille, which often cause accessibility and affordability issues (Sadak et al., 2024). Traditional methods of instruction for visually impaired students require a lot of time and effort, and need the attention and hard work of teachers. Supalkar and Naik (2021) believed that due to the challenges of

learning Braille using traditional methods and lack of trained instructors, the literacy of visually impaired students is extremely poor.

According to Rahmat et al. (2024), most people in Indonesia are not proficient in Braille as it is not a mandatory subject in the country's educational system creating a slight communication gap between people with and without visual impairments. Even though Information and Communication Technology (ICT) has impacted every industry, there is still a gap between visually impaired people due to ignorance and inattention. Jariwala and Popat (2025) believed that people with visual impairments cannot communicate because of constraints of tools and resources.

Despite the attempt of higher education institutions to make necessary accommodations for students with visual impairments, institutions did not completely succeed in meeting students' needs in accessing some assessment practices, including some students' inability to read braille grade 2 and insufficient technology skills (Lawi & Muzata, 2025). Therefore, learning Braille is not a simple process, especially for the general populace impairments. While visualizations turn raw data into understandable insights, they frequently exclude visually impaired communities. Despite the availability of tools such as screen readers and alt texts to translate visual information, visual literacy is still a major barrier to visual data interpretation (He & Yu, 2024).

Visually impaired students encountered various challenges such as lack of material, difficulty accessing lecture notes, not getting much academic support,

insufficient academic resources, and not having access to the universities (Firat, 2021).

### **Review and Related Studies**

Several developers have attempted to create and design electronic braille devices addressing the needs of visually impaired students in an educational context.

With U.S. Patent no. US9171483B2, this system is for visually impaired individuals. It has a learning information server, along with a lecture and authoring program, and a first output device (PC), which receives learning information from a learning information server via a wired/wireless network (LAN or WLAN), and has a viewer program installed therein for enlarging and reducing the screen function. A second information output device (haptic electronic board) comprising a wireless network function (ZigBee or Bluetooth) for transferring the information to the learning information server, unique device ID for accessing the learning information server through a local wireless network (ZigBee or Bluetooth), a function for converting the learning information received from the learning information server into haptic information, and a function for transferring the haptic information to the blind people.

Furthermore, the device with a Korean Patent no. KR20250066954A, addresses the issues of visually impaired students in educational context. It generates a content step for learners with a multi-layer feature. However, customized learning content can also be provided to learners by considering

several factors such as degree of disability, intellectual capacity, etc. of visually impaired learners in online and offline education sites using tactile displays, mobile terminals, etc.

Moreover, an invention with a technique using a computer to dynamically modify the virtual Braille keyboard on touch screens was introduced, in which multiple virtual keys are associated with different finger positions. The method comprises the steps of detecting typing events, each of which involves one or more characters being typed. Identifying, at one or more touch locations, one or more fingers of the user that are in contact with the touch screen during the typing event and identifying one or more active fingers during the typing event (DR. Rambo et al., 2017).

Reid et al. (2014) invention allows a user to concurrently present a primary content item and its accompanying braille translation may be accessible through a computing device. An e-book, audiobook, or other type of media could be the principal content item. Character, place, and image descriptions, endnotes, footnotes, glossaries, appendices, author, editor, or other commentary, and other supplemental information may be included with the primary content item. Additionally, there may be a braille translation attached to the supplemental information.

Moreover, a device was invented to help visually impaired students study braille and create art in both remote and classroom settings when used in conjunction with a computer or smartphone software (Lubiner, 2022).

Similarly, Dawle et al. (2019) developed a smart braille learning designed to enhance the educational experience of the visually impaired learners and offers a platform for listening to audio-braille tactile lessons that incorporate speech and braille data input or output. The smart interactive system includes a digital braille 10-slate, a braille keyboard, a speaker for haptic and audio data output, a big braille display and a normal braille display, a proximity touch sensor for input, and a vibration motor. The digital braille slate provides both auditory feedback through a speaker and mechanical tactile response when the stylus is used to press each braille dot. Since the system is online, it is possible to update software and content remotely and upload usage analytics data.

Furthermore, an innovation was developed, which comprises a terminal, system and braille display module. It provides a communication module, control module and a braille display module for receiving, processing and converting information into braille for the benefit of visually impaired students allowing them to access the information that they need immediately (Chen et al., 2017).

Another innovation, with U.S patent no.US20250181696A1, comprises a processor and a non-transitory computer readable medium that is coupled to the processor. In some embodiments, the non-transitory computer-readable medium contains code that generates a randomized set of keys for authentication purposes, communicates to a reconfigurable braille (RB) board to render raised RB board keys on the RB board according to a randomized order of the randomized set of keys, and utilizes the raised RB board keys to ascertain an authentication code that is utilized to authenticate a user of a mobile device.

Additionally, a braille generation device that was invented includes a user-held housing, an input module positioned on the front and back sides of the housing that provides an input area for receiving a braille input from the user, and a control module that sets a functional area in a predetermined area on the input area on the back side of the housing, sets a plurality of braille element areas in the input area on the front side of the housing according to a predetermined arrangement, and generates braille based on the braille input for the plural braille element areas and the functional area sensed by the input module (Jo Don-woo, 2019).

Building on this, Yu Chung-gi (2024) developed a device that comprises a use terminal for scanning a braille block, and a location recognition server for providing a straight line notification or a direction change notification corresponding to the scanned braille block.

Moreover, a braille assistance device was developed to display and create braille using telescoping salient points that are managed by an actuator. Word information is captured by an image device and then converted into braille by the control unit before being displayed. It can be incorporated into laptops, tablets, or phones to make reading more convenient for blind people (Xiào ào, 2018).

Furthermore, Horbal (2017) invented a system that allows blind or visually impaired people to find typical touch locations on a device through the use of an assistive grid that overlays on the touchscreen. It supports usability movements,

offers a standardized user interface for websites and apps, and could include raised Braille for direction.

Additionally, an innovation offers electromagnetic Braille displays and other tactile displays that utilize detachable capture plate assemblies designed for easy removal during cleaning and maintenance (Moon et al., 2021).

Similarly, Californiaa (2021) invented a self-scrolling Braille pad for output and a chordic keyboard for input, allowing deafblind people to read, write, and communicate. Duplex communication is supported, hands can stay in position, and Braille reading is made easier by eliminating the need for swiping motions.

Another device was created that uses the Up and Down Mechanism. A Braille display is provided employing electrowetting forces to control the sequential raising and lowering of the pins. Moreover, this device provides a method of interactive reading whereby the reading of Braille from the display is computer synchronized with other processes to help users learn braille and increase the user's experience.

Building on this. Moon et al. (2020) proposed an electromagnetic refreshable Braille reader that includes an actuation mechanism comprising a permanent magnet for latching and a coil adjacent to the magnet. Additionally, the system may include a separate enclosure with detachable tactile components that users can easily remove and clean.

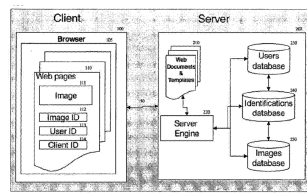
Moreover, a braille block was introduced consisting of a base substrate, multiple braille units, and a low-frequency supplying unit. The braille units are formed on the base substrate, each comprising six points. The low-frequency supplying unit supplies low-frequency current to each of the braille units to allow a user to distinguish braille letters by low-frequency electrostimulation (Yong Hwang-gu, 2014).

Similarly, Ramírez (2018) invented a device for Braille writing comprising a casing with a front side and a top side adjacent to the front side, where a processing unit is placed inside the casing. The device also has a button-positioning mechanism located on the front of the casing and connected to the front buttons. This positioning mechanism allows changing the position of each front button.

Furthermore, an electronic Braille Interface was developed as a low-cost portable device that uses moveable pins which are controlled by springs and magnets to convert ASCII text into physical Braille. This allows blind people to read from e-readers, laptops, and phones (Porfirio, 2019).

A system innovated by Lee Seok (2025) introduced a braille output plate with moving dot rods controlled by actuators allowing for dynamic braille creation. Then, the variable braille output device transforms input data (picture, text, audio, video, or motion) into braille codes before displaying them.

## State-of-the-art Landscape

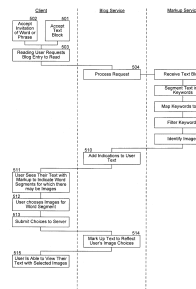


**Inventors:** Mark Frederick Frigon

**Date patented:** 2019-11-28

**Classification number:** G06F16/583

**Publication number:** US20190361931A1

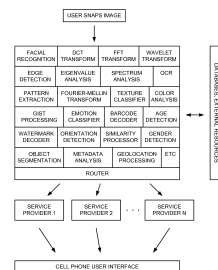


**Inventors:** Steven F. Heck

**Date patented:** 2016-01-05

**Classification number:** G06F17/30554

**Publication number:** US9229992B2

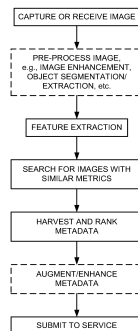


**Inventors:** Tony F. Rodriguez

**Date patented:** 2016-02-23

**Classification number:** H04W4/16

**Publication number:** US9271133B2

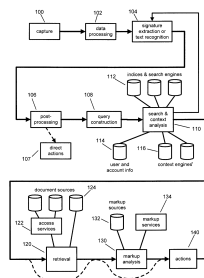


**Inventor:** William Y. Conwell

**Date patented:** 2015-08-11

**Classification number:** G06K9/0063

**Publication number:** US9104915B2



**Inventor:** Martin Towle King, Dale L. Grover, Clifford A. Kushler, James Quentin, Stafford-Fraser

**Date patented:** 2013-05-21

**Classification number:** G06F16/951

**Publication number:** US8447144B

Figure 2. State of the Art Landscape of the Study

## PRIOR ARTS

### Prior Arts 1

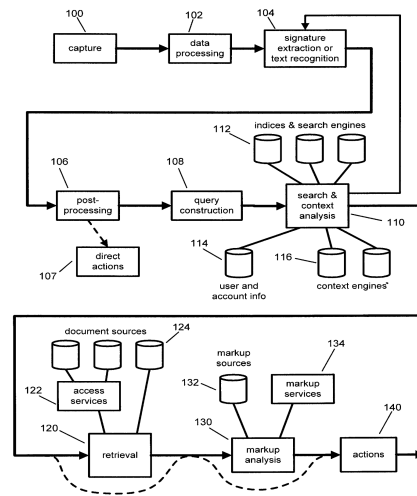


Figure 3. Data Capture

Inventor: Martin Towle King, Dale L. Grover, Clifford A. Kushler, James Quentin, Stafford-Fraser

Date patented: 2013-05-21

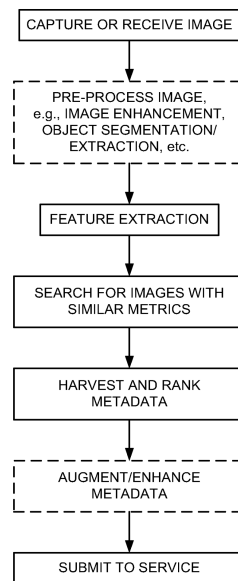
Classification number: G06F16/951

Publication number: US8447144B

### Description

It is described as a portable device with scanning, imaging or other data-capture capabilities. In certain embodiments, the device can notify a user when sufficient information has been captured to uniquely identify a source document. The device can also produce timestamps and location-stamps to capture when and where the data was captured. In other embodiments, the device is controlled by user gestures. The portable device might also store billing data and be used to process content or service subscriptions related to its usage.

## Prior Arts 2



*Figure 4. Content Processing*

Inventor: William Y. Conwell

Date patented: 2015-08-11

Classification number: G06K9/0063

Publication number: US9104915B2I

## Description

A handheld device, which consists of a camera and a microphone, records images and audio that depict an item. The device identifies a single or multiple descriptors associated with the object through analysis of the captured imagery and extracts descriptors associated with the object based on other imagery or audio sources. The device then uses one or more electronic processors to evaluate the descriptors to determine whether the object is probably of a first category, a second category, or a third category. Depending on the classification that the device arrives at, the device will take one or more actions based on the classified class of the object.

### Prior Arts 3

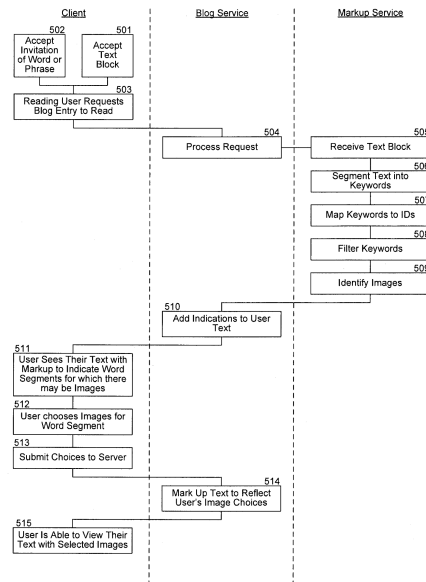


Figure 5. Digital Content Identification

Inventor: Steven F. Heck

Date patented: 2016-01-05

Classification number: G06F17/30554

Publication number: US9229992B2

### Description

An indication of one or more words in a block is fed to a system which identifies digital content related to a part of a block of text. The system works with a database and finds out digital content items or identifiers related to the words indicated. The retrieved objects are then shown to the user, who then chooses the objects and the system matches the chosen objects with the words shown or played. Other incarnations of the system are revealed as well.

## Matrix of Novelty

**Technical Field:** Academic Field

| Disclosed Features                                                                               | Prior Arts Features |    |    |
|--------------------------------------------------------------------------------------------------|---------------------|----|----|
| <b>Proposed Project:</b><br>Braille Connect: A Mobile Application for Visually Impaired Students | S1                  | S2 | S3 |
| Braille-to-Word/Speech Translation                                                               | ×                   | ×  | ×  |
| Talkback Description                                                                             | ×                   | ✓  | ×  |
| File Upload Capability (.pdf, .docs, .txt)                                                       | ×                   | ×  | ×  |
| Assessment Mode                                                                                  | ×                   | ×  | ×  |
| Scanned object/content Identification                                                            | ✓                   | ✓  | ✓  |

*Table 1. Matrix of Novelty*

Table 1 shows a comparison of some of the features of developed technology. It has similarities with other closest prior arts; however, there are differences in the qualities that make the proposed project distinctive and innovative.

## Claims

The researchers claim the following functionality and features of the "Braille Connect: A Mobile Application for Visually Impaired Students":

1. Braille-to-word translation
2. File upload capability such as PDF, DOCS, and TXT
3. Assessment Mode

## Synthesis

The examined preceding arts outline different technological methods that can lead to easy and smart systems, specifically in data capture, object recognition, and content interaction. One system is a portable device that has scanning, imaging, and data-capture features that can be used to know when enough information has been collected to recognize a document. It also includes support of features like timestamping, location tracking, and gesture-based control allowing the efficient and context-sensitive data processing (US8447144B).

Another predecessor art presents a handheld device that has a camera and microphone functionality, allowing it to take pictures and audio of objects. The device uses electronic processors to obtain the descriptors and categorize objects using data analysis. According to the classification outcome, the system carries out appropriate actions, which show smart object recognition and response systems (US9104915B2).

Also, a system is introduced that takes the chosen words of a block of text and finds the corresponding digital information in a database. The content is retrieved and presented to the user, who is able to select the relevant items, and the system compares the selections with the relevant words. This method emphasizes the interactive content retrieval and context connection of text and digital materials (US9229992B2).

Together, the preceding arts reflect progressions in the spheres of data retrieval, object recognition, and interactive content systems. Nevertheless, although they offer important functionalities, they are not specifically targeted at the needs of the visually impaired learners but are targeted at general-purpose applications. This presents the possibility of further innovation, especially in terms of unifying these technologies into a single, accessible platform that will facilitate translation of Braille, audio feedback, and delivery of educational content to the needs of visually impaired students.

### **Theoretical Framework**

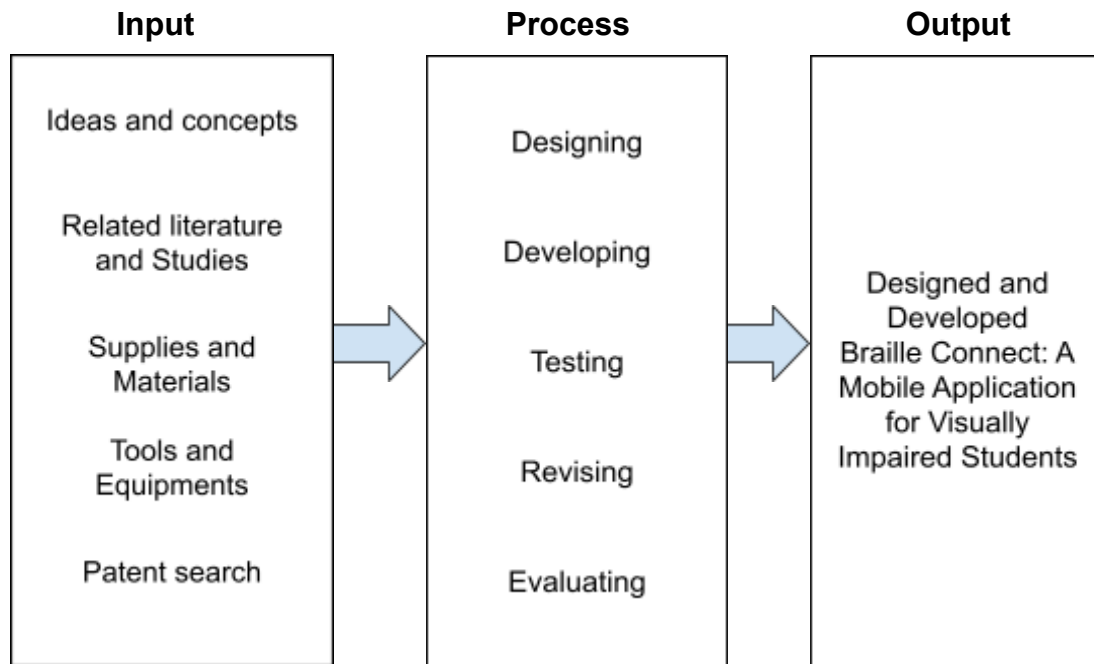
This section presents the two core theories that support the present study. The first theory is based on Assistive Technology Theory, which emphasizes that technology serves as a support to compensate for functional limitations, support skill development, and promote inclusion among learners with disabilities, especially learners who are visually challenged. Assistive technology is specified as any equipment or system that maintains, increases, or improves functional capabilities that allow students to participate more fully in learning and daily activities (World Health Organization: WHO, 2024; Assistive Technology Theory - 1373 Words | Bartleby, n.d.-b). It functions not only as a compensatory tool but also as a developmental and inclusive support, ensuring equitable access to education and opportunities (Cook & Polgar, 2019). For this purpose, the assistive technology theory offers a guiding framework to examine how tools and

applications can enhance the accessibility and engagement of students with visual impairments.

The second theory is the Theory of Inclusive Education, which emphasizes that all learners should be given equal opportunities to participate meaningfully in mainstream classrooms, regardless of disability or background. Ainscow (2005) suggests adapting teaching methods, removing barriers, and fostering environments where diversity is recognized and strengthened. Inclusion should ensure active participation, equitable access, and a sense of belonging rather than physical designation in schools. It presents a perspective to review how school institutions can potentially develop learning environments that facilitate visually challenged students while promoting equity and social justice.

Both theories marked that the effective integration of educational innovations requires balanced technological solutions with inclusive pedagogical practices and human-centered values. They uncover that through feedback, adaptation, and continuous refinement, consequently, teaching methods can evolve. Nevertheless, institutional frameworks, inclusive strategies, and clear policies that align technology use with the wider goal of equity must support such progress. As a result, this research innovation initiative could be advantageous toward addressing the educational challenges of students who are visually challenged, since it enables school institutions to handle learning resources more effectively while assuring participation, accessibility, and a sense of inclusion for all learners.

## Conceptual Framework



*Figure 6. Conceptual Model of the Study*

The conceptual framework of the study was based on the most commonly used model in most of the studies and project developments that involve the creation of a tangible technology to advance human efforts, namely, Input-Process-Output model. The conceptual model is shown in block diagram form in Figure 6. The inputs contained by the ideas, related literature, supplies, materials, tools, equipment, patent search, references, and the researcher's personal ideas and experiences are all included in the Input Block. The Process Block was the design and development of the proposed process that made for excellent output for body parts that were to be durable. Then, the efficiency of the technology's time and functionality is tested and the defect is revised based on the test. Finally, evaluating the output for its significance should be done before

the presentation of the output. The Output Block, which is the designed and developed Braille Connect Mobile Application, is the third part of the block diagram.

All of these components were used in the design and development of the study. The input block's components were refined and highlighted the researcher's thoughts, thereby acting as a guide and continuum for the actualization of the study. To produce a research product in the form of a technology, the researcher diligently completed the activities listed in the process block. Lastly, the statement in the third block illustrates the primary objective of the study, which is to create the Braille Connect Mobile Application, a tangible product.

## **Operational Definition of Terms**

This section presents the operational definition of terms related to the project as follows:

### **Braille-to-Word/Speech translation**

Braille-to-word/speech

translation is the translation of Braille characters into readable words or speech. The technology allows the visually impaired to comprehend the input of the Braille using a digital text display or text-to-speech output.

### **Mobile Application**

A software program specifically designed to run on handheld devices like smartphones and tablets.

**Scanned object/content Identification** Scanned object/content identification is a technology that scans on cameras or sensors to record images or data and processes to identify objects, text or content. The system interprets the input it captures and gives its user valuable information about the identified item.

**Talkback Description** Talkback description is an accessibility feature that allows users to receive audio feedback by turning text, buttons, and actions on the screen into audio. It enables blind people to navigate and interact with digital devices using voice prompts.

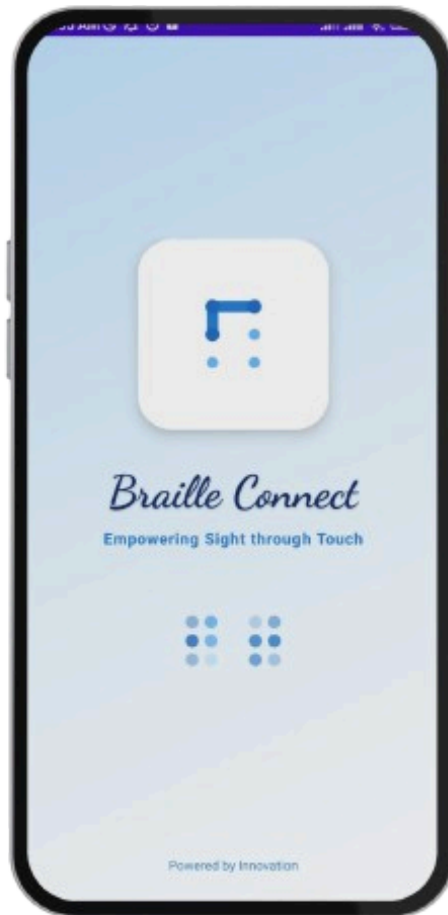
## Chapter III

### DEVELOPMENT AND EVALUATION OF THE PROJECT

#### Project Design Phase

This displays the study's concept design, required measurements, materials and supplies, instruments, and tools.

#### Concept Design



*Figure 7. Loading Screen*

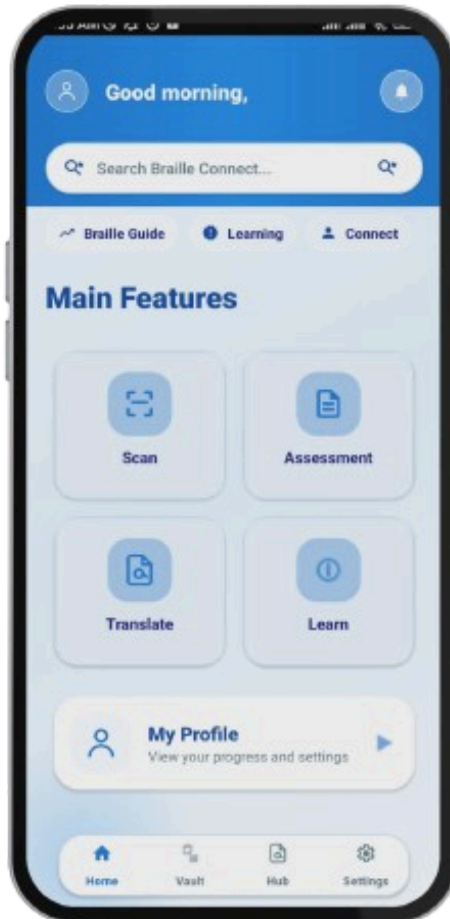
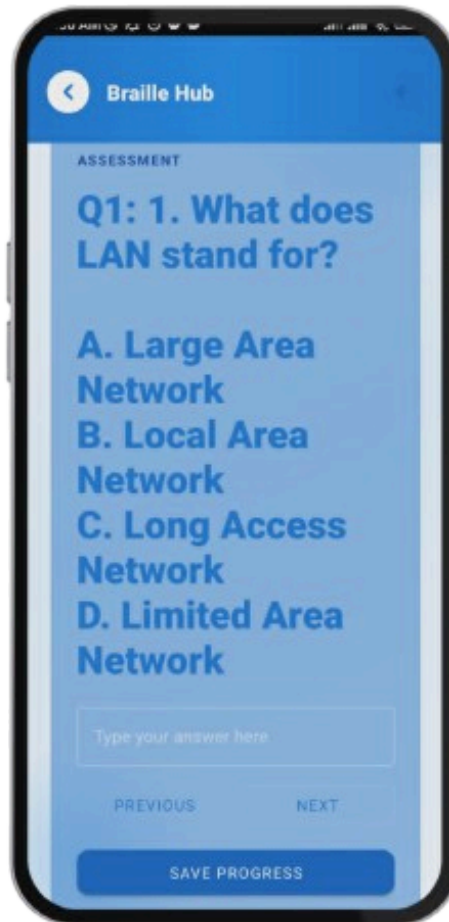


Figure 8. Homepage



*Figure 9. Assessment Page*

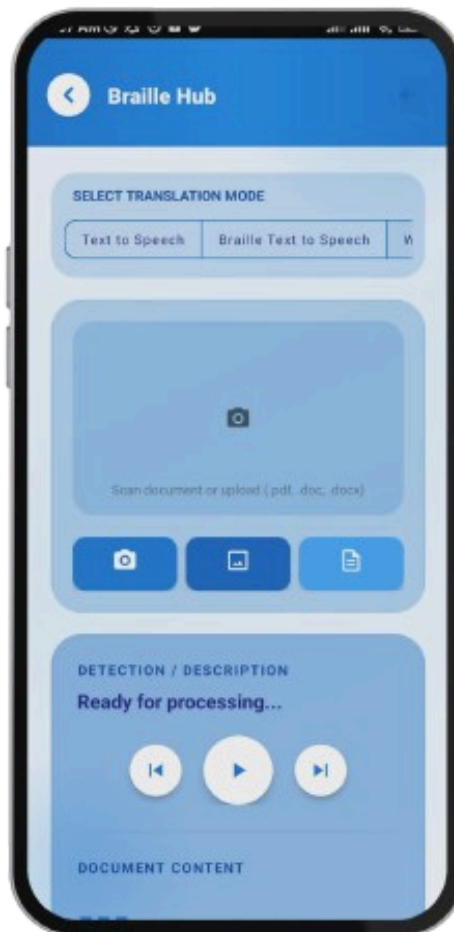


Figure 10. Translate Page



Figure 11. Scanner Page



Figure 12. Profile Page

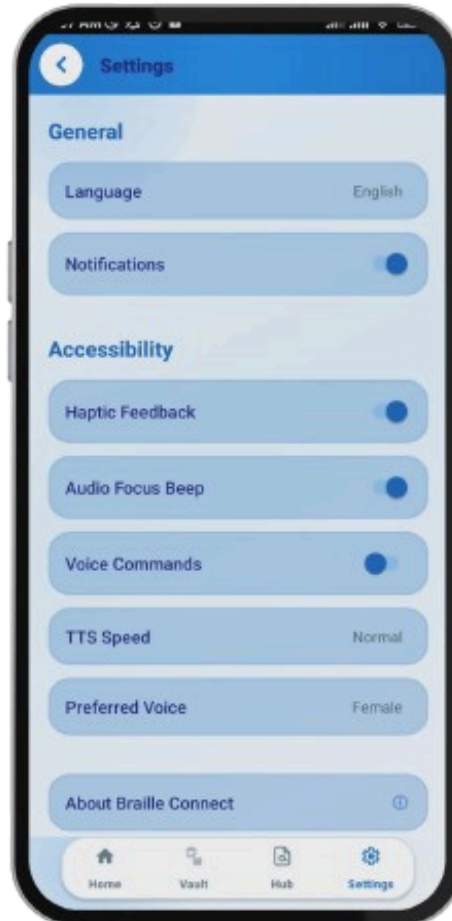
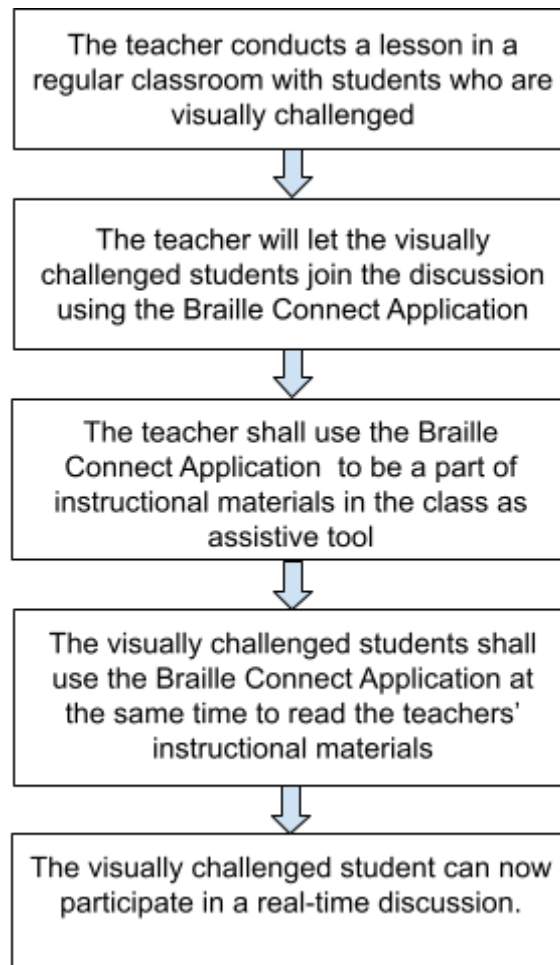


Figure 13. Settings

## New Process Flow



*Figure 14. New Process Flow*

The teaching process will be initiated by the instructor teaching in a typical classroom with both the normal and the visually impaired students. This will encourage a non-segregated space of learning whereby all learners will share in the learning process without being segregated on the basis of visual ability.

In the discussion stage, the teacher will enable visually challenged students to actively participate via the Braille Connect Application. This makes

sure that they are not excluded in interaction in the classroom and are able to follow the lesson.

The teacher then incorporates the Braille Connect Application in the instructional resources and employs it as an aid to convey content. Lesson materials are also prepared using this system and presented in a format that is readable by visually challenged learners.

The visually challenged students follow the Braille Connect Application as the lesson goes on, and they read the instructional materials given by the teacher at the same time as the rest of the class. Such real-time access will help them to follow the lesson.

This means that the visually challenged students can actively engage in real-time discussions, thus improving engagement, accessibility, and equal learning opportunities in the classroom.

## Specification of the Project

|                           |                            |
|---------------------------|----------------------------|
| App Name :                | Braille Connect            |
| App Size :                | 130 MB                     |
| Supported OS :            | Android                    |
| Required OS :             | Android 7.0 and above      |
| Minimum RAM :             | 6GB RAM in Android phones  |
| Permission :              | Storage, Camera, and Files |
| Maintenance and Updates : | By patch                   |

## Software Tools and Device

These software tools and devices will be used to conduct the study.

These are all related to software creation.

| Name                            | Description                                                              |
|---------------------------------|--------------------------------------------------------------------------|
| <b>Android Studio / VS Code</b> | Used for coding and developing the mobile application.                   |
| <b>Firebase (Free Tier)</b>     | Used for database management and user authentication.                    |
| <b>Laptop / PC</b>              | Used for programming, development, and testing of the application.       |
| <b>Android Smartphone</b>       | Used for installing and testing the mobile application on a real device. |

Table 2. Software Tools and Devices

## **Development Phase**

This section describes the development techniques, as well as the project's construction timeline, development tools, and device costs.

## **Construction, Assembly and Fabrication Procedure**

### **1.) Designing and Developing Plan**

1.1 Draft the design

1.2 Illustrate the user interface of the software application

### **2.) Developing the Application.**

2.1 Gather the device needed.

2.2 Download all necessary development tools

2.3 Create the software application

### **3.) Pilot-testing of the device.**

4.1 Operate/Utilize the application

4.2 Test its functionality within a week.

### **4.) Revise the faulty functionality of the application.**

5.1 Fix the errors or bugs in the software application

5.3 Test its functionality again.

### Construction Time Frame

| Task                                                        | Month 1 |  |  | Month 2 |  |  | Month 3 |  |  | Month 4 |  |  | Month 5 |  |  |
|-------------------------------------------------------------|---------|--|--|---------|--|--|---------|--|--|---------|--|--|---------|--|--|
| 1. Plan and draft app features                              |         |  |  |         |  |  |         |  |  |         |  |  |         |  |  |
| 2. Research and gather data from related studies            |         |  |  |         |  |  |         |  |  |         |  |  |         |  |  |
| 3. Design UI/UX                                             |         |  |  |         |  |  |         |  |  |         |  |  |         |  |  |
| 4. Set up database and backend                              |         |  |  |         |  |  |         |  |  |         |  |  |         |  |  |
| 5. Code main features                                       |         |  |  |         |  |  |         |  |  |         |  |  |         |  |  |
| 6. Test app functionality                                   |         |  |  |         |  |  |         |  |  |         |  |  |         |  |  |
| 7. Debug, assess, and revise errors                         |         |  |  |         |  |  |         |  |  |         |  |  |         |  |  |
| 8. Usability testing with selected respondents              |         |  |  |         |  |  |         |  |  |         |  |  |         |  |  |
| 9. Final presentation and submission of Braille Connect app |         |  |  |         |  |  |         |  |  |         |  |  |         |  |  |

Table 3. Construction Time Frame

## Costing

The table shows the costing of supplies and materials. This all means of expenditure and budget appropriation of the project with the 1.5% price inflation.

| Category             | Description                        | Estimated Cost (₱) |
|----------------------|------------------------------------|--------------------|
| Development Tools    | Free IDEs (Android Studio/VS Code) | ₱ 0.00             |
| Testing Devices      | Personal phones/laptops            | ₱ 0.00             |
| Internet/Data        | WiFi or load contribution          | ₱ 1,500.00         |
| <b>Total Costing</b> |                                    | <b>₱ 1,500.00</b>  |

*Table 4. Costing*

## Cost and Expenditures

| Expenditures           | Cost              |
|------------------------|-------------------|
| Total Development Cost | ₱ 1,500.00        |
| Labor Cost             | ₱ 0.00            |
| Fabrication Cost       | ₱ 0.00            |
| <b>Total Cost</b>      | <b>₱ 1,500.00</b> |

*Table 5. Cost of the Software Application*

## Pilot Testing and Revision of the Project

| Defect Found | Revision Made |
|--------------|---------------|
| <i>None</i>  | <i>None</i>   |

*Table 6. Pilot Testing and Revision*

## Evaluation Phase

This section covers the research design and instrument of the study, the evaluators who will evaluate this study, the data collection techniques, and the stats needed for the project's creation.

## Testing and Revision Phase

After the application has been developed, it will be tested to ensure its functionality; any problems discovered and revisions required will be displayed in the table once the testing and revision phase has been completed.

## Research Design

### Research Instrument

The research study collected data using survey questionnaires. The instruments utilized to collect data for this investigation are as follows:

**Semi-structured Interview.** The respondents were asked about their ease of use and overall learning experience with the new mobile application.

**Survey Questionnaire.** The questionnaire served as the major primary research method for gathering initial hard data. The questionnaires were delivered to the respondents.

The respondents were given a survey instrument along with a preference or agreement scale. The study made use of the Likert functional scale. The mean rating was as follows:

5 (4.50-5.00) - Highly Functional/Usable/Effective (HF/HU/HE)

4 (3.50-4.49) - Functional/Usable/Effective (F/U/E)

3 (2.50-3.49) - Moderately Functional/Usable/Effective (MF/MU/ME)

2 (1.50-2.49) - Slightly Functional/Usable/Effective (SF/SU/SE)

1 (1.0-1.49) - Not Functional/Usable/Effective (NF/NU/NE)

## **Evaluators**

The researchers used stratified purposive sampling to determine the study's results, which consisted of 5 students (from all grade levels), 5 parents, and 10 mainstream teachers who interact with blind students. The researchers explained the application's purpose and features, demonstrated real-time lessons or instruction, and then allowed each respondent to use the device and assess its usability and acceptability by answering a survey questionnaire and conducting a brief semi-structured interview. The data were collected and evaluated to produce a data analysis that includes the Likert Scale and a recommendation tick box.

## **Sampling Technique**

The study employed non-probability sampling, particularly purposive sampling. Purposive sampling allows researchers to gain a deeper knowledge of complicated phenomena by consciously selecting specific aspects based on their relevance to the study subject (Tajik et al., 2025).

## **Data Gathering Procedure**

The following were the procedures performed in conducting the study:

1. Seek permission from the College President and Program Head to conduct a survey outside the City College of Davao.
2. Ask permission from the Principal of Davao School for the Blind to conduct the survey.

3. Conduct a survey to 10 Special Education Teachers or regular teachers who interact with visually challenged students, 5 students across grade levels, and 5 parents with visually challenged children.
4. Gather the data and subject the result to the statistician for interpretation.

### **Statistical Treatment**

Upon gathering the data, the mean of the results was assessed and analyzed.

### **Mean / Average**

The mean was calculated, and the weighted average was used to estimate the equivalent of the responses provided. This computation is important for determining how effective and functioning the device is.

**MEAN** – is the average score. Numerically, it is equal to the sum of the scores over the numbers of scores.

Formula:

Where:  $\bar{x}$  = Mean,  $\sum X$  = sum of responses, and  $N$  = total number of responses.

## CHAPTER IV

### RESULTS AND DISCUSSION OF EVALUATION AND DESCRIPTION OF THE COMPLETED PROJECT

#### Presentation of Data

This section details the results of the evaluation of the Braille Connect Application's functionality, usability, and effectiveness from our questionnaire. The following descriptive ratings are based on a 5-point Likert Scale, which is:

*5 (4.50-5.00) - Highly Functional/Usable/Effective (HF/HU/HE)*

*4 (3.50-4.49) - Functional/Usable/Effective (F/U/E)*

*3 (2.50-3.49) - Moderately Functional/Usable/Effective (MF/MU/ME)*

*2 (1.50-2.49) - Slightly Functional/Usable/Effective (SF/SU/SE)*

*1 (1.0-1.49) - Not Functional/Usable/Effective (NF/NU/N)*

| <b>Questions for Functionality</b>                                                         | <b>Mean</b> | <b>Descriptive Rating</b> |
|--------------------------------------------------------------------------------------------|-------------|---------------------------|
| 1. When scanning the surroundings, the application captures images clearly and accurately. | <b>4.8</b>  | <b>Highly Functional</b>  |
| 2. The system accurately translates word-to-braille.                                       | <b>4.85</b> | <b>Highly Functional</b>  |
| 3. The application generates clear and easy-to-understand image descriptions.              | <b>4.6</b>  | <b>Highly Functional</b>  |
| 4. The scanning feature processes images quickly and efficiently.                          | <b>4.85</b> | <b>Highly Functional</b>  |
| 5. The TalkBack feature produces clear and audible voice descriptions.                     | <b>4.8</b>  | <b>Highly Functional</b>  |
| <b>Grand Mean</b>                                                                          | <b>4.78</b> | <b>Highly Functional</b>  |

*Table 7. Results of Functionality Test*

| Questions for Usability                                                                                      | Mean        | Descriptive Rating   |
|--------------------------------------------------------------------------------------------------------------|-------------|----------------------|
| 1. The application interface is accessible for visually impaired users.                                      | 4.7         | Highly Usable        |
| 2. The controls of the application are easy to understand and operate.                                       | 4.6         | Highly Usable        |
| 3. The scanning feature can be used without difficulty by blind students.                                    | 4.65        | Highly Usable        |
| 4. The audio output of the application is clear and easy to follow.                                          | 4.75        | Highly Usable        |
| 5. The application functions without lag, crashes, or performance issues during extended and continuous use. | 4.55        | Highly Usable        |
| <b>Grand Mean</b>                                                                                            | <b>4.65</b> | <b>Highly Usable</b> |

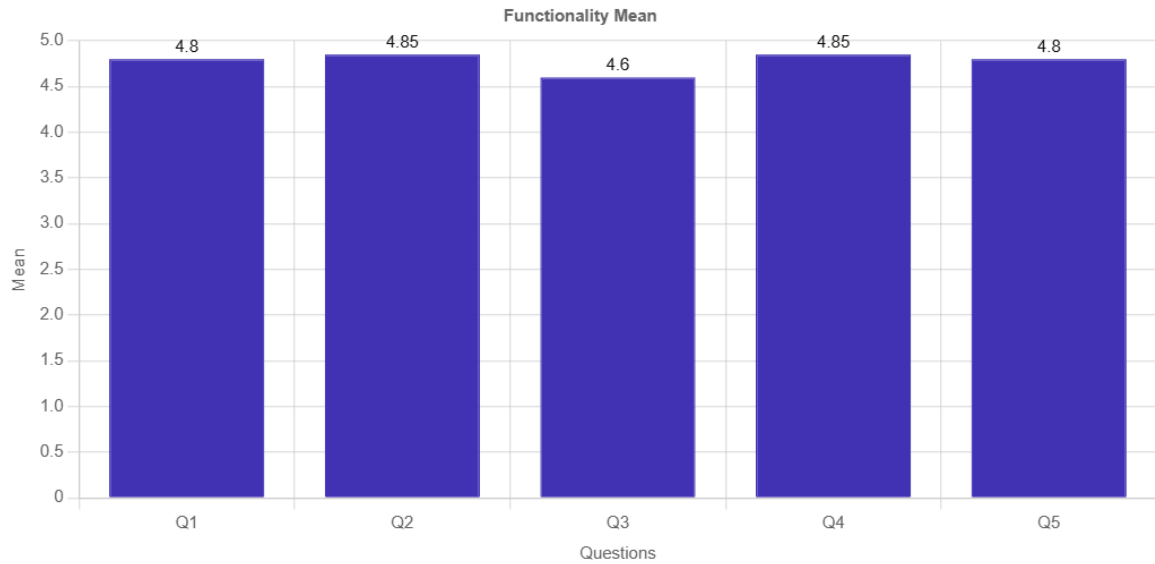
*Table 8. Results of Usability Test*

| <b>Questions for Effectiveness</b>                                                       | <b>Mean</b> | <b>Descriptive Rating</b> |
|------------------------------------------------------------------------------------------|-------------|---------------------------|
| 1. The application helps visually impaired users identify objects in their surroundings. | <b>4.65</b> | <b>Highly Effective</b>   |
| 2. The application supports visually impaired students in their learning activities.     | <b>4.85</b> | <b>Highly Effective</b>   |
| 3. The system assists teachers and parents in guiding visually impaired students.        | <b>4.75</b> | <b>Highly Effective</b>   |
| 4. The application improves the user's awareness of their environment.                   | <b>4.75</b> | <b>Highly Effective</b>   |
| 5. The application contributes to the independence of visually impaired users.           | <b>4.55</b> | <b>Highly Effective</b>   |
| <b>Grand Mean</b>                                                                        | <b>4.71</b> | <b>Highly Effective</b>   |

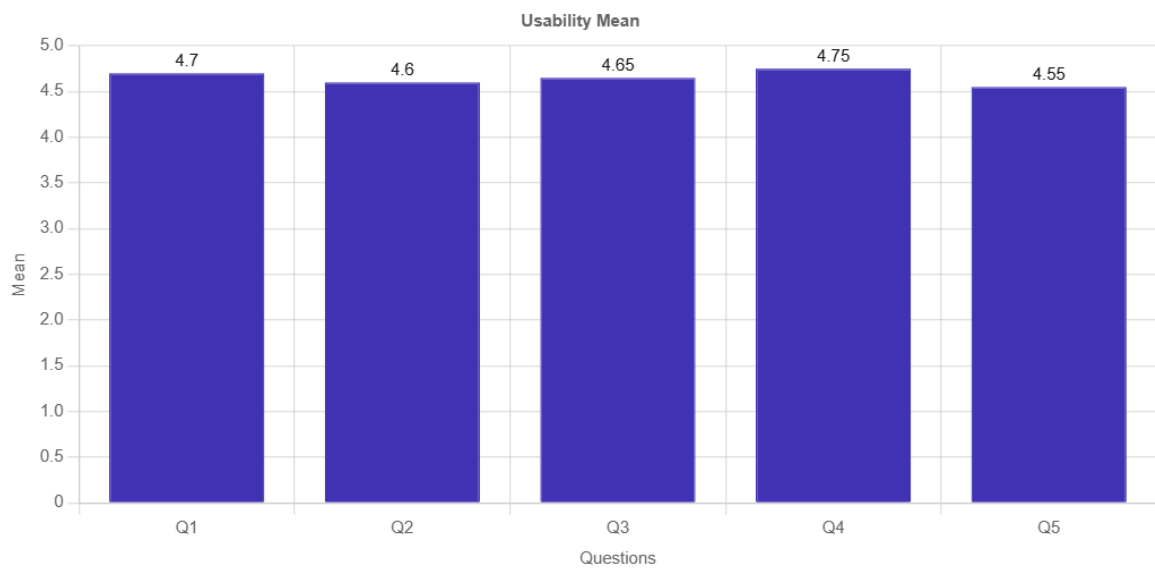
*Table 9. Results of Effectiveness Test*

## Analysis of Data

Presentation of data and result discussion of the project evaluation.



*Figure 15. Bar Graph for Functionality Mean*



*Figure 16. Bar Graph for Usability Mean*

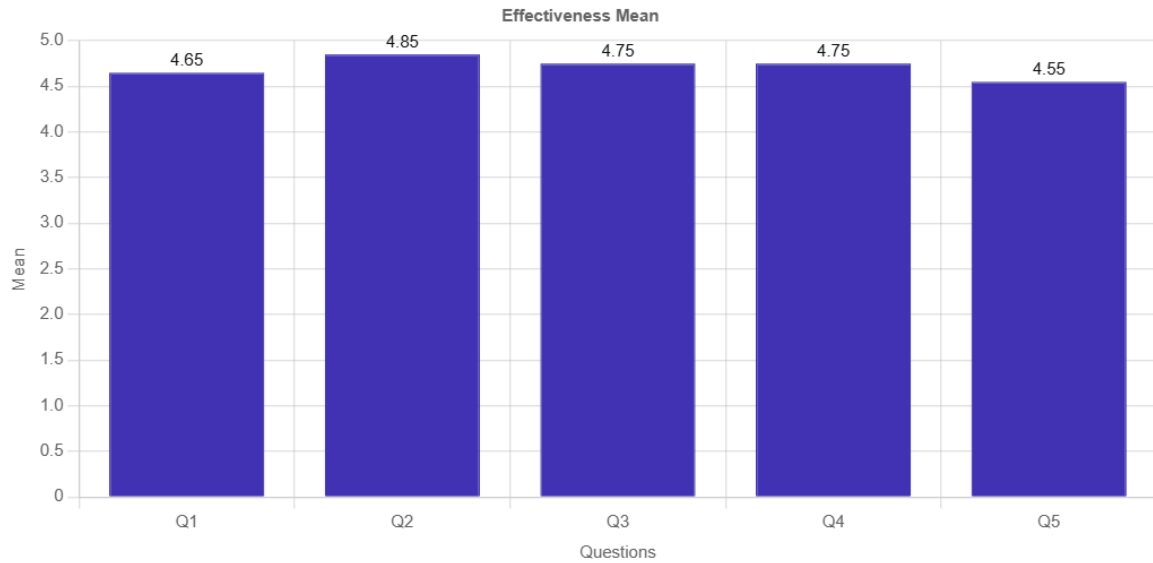


Figure 17. Bar Graph for Effectiveness Mean

| Rating Scale | Rating      | Descriptive Rating                     |
|--------------|-------------|----------------------------------------|
| 5            | (4.50-5.00) | Highly Functional/Usable/Effective     |
| 4            | (3.50-4.49) | Functional/Usable/Effective            |
| 3            | (2.50-3.49) | Moderately Functional/Usable/Effective |
| 2            | (1.50-2.49) | Slightly Functional/Usable/Effective   |
| 1            | (1.0-1.49)  | Not Functional/Usable/Effective        |

Table 10. Likert Scale for Functionality, Usability, and Effectiveness Test

### **Interpretation of Data for Functionality**

Based on the data in Table 7 above, it shows that the Braille Connect Mobile Application has obtained an overall mean score of 4.78, which is interpreted as Highly Functional. Both statements *The system accurately translates word-to-braille* and *The scanning feature processes images quickly and efficiently* got the highest mean of 4.85, with a description of Highly Functional. This implies that the Braille Connect application performs its intended functions accurately and efficiently. On the other hand, the statement that got the lowest mean is *The application generates clear and easy-to-understand image descriptions*. The mean score is 4.60, with a description of Highly Functional level. *When scanning the surroundings, the application captures images clearly and accurately* and *The TalkBack feature produces clear and audible voice descriptions* scored the same mean (4.8) which indicates that the application is Highly Functional in terms of the accuracy and clearness of captured images, and audible voice descriptions in the TalkBack Feature. The Braille Connect Mobile application was tested using Twenty (20) respondents in different groups.

### **Interpretation of Data for Usability**

The data in Table 8 shows that the Braille Connect Mobile Application gained an overall mean score of 4.65, interpreted as a Highly Usable application based on the Twenty (20) respondents from different groups. The statement that gained the highest mean is *The audio output of the application is clear and easy to follow* with the mean score of 4.75 rated as Highly usable. On the other hand,

*The application functions without lag, crashes, or performance issues during extended and continuous use* obtained the lowest mean score (4.55), which indicates that the application functions with some minor performance issues during continuous use, though it still falls within the Highly Usable Level. The statement *The application interface is accessible for visually impaired users* garnered a mean of 4.7, with a description of Highly Usable. This means that the Braille Connect Mobile Application's interface is accessible for visually impaired users. The results still show consistent High descriptive ratings in terms of the Application's Usability.

### **Interpretation of Data for Effectiveness**

The data in Table 9 displays that the Braille Connect application acquired an overall mean score of 4.71, interpreted as Highly Effective. The statement that got the highest mean is *The application supports visually impaired students*. The mean score is 4.85, with a description Highly Functional. This explains that the application is highly effective in terms of supporting visually impaired users in their learning activities. Whereas, *The application contributes to the independence of visually impaired users* received the lowest mean score (4.55), though still rated within the highly effective level. Both statements *The system assists teachers and parents in guiding visually impaired students* and *The application improves the user's awareness of their environment* garnered the same mean score (4.75), which indicates that the application is highly effective with regard to improving visually impaired users' awareness of their environment

and assisting teachers and parents in guiding visually impaired users.

Overall, the results in functionality, usability, and effectiveness showed consistent ratings. This means that the application is functional, effective, and user-friendly in handling visually impaired users.

### **Findings of the Study**

The Braille Connect Mobile Application was successfully designed and developed to support visually impaired students in their education, promoting mainstream learning. The application obtained an overall functionality, usability, and effectiveness mean of 4.78 (Highly Functional), 4.65 (Highly Usable), and 4.71 (Highly Effective). This indicates that all of the objectives were achieved, as the application effectively allows the visually impaired students to translate Braille text to Word and Word to Braille accurately, navigate the user interface efficiently, and improve the user's awareness of their environment.

The data also shows that the application received high mean scores in terms of functionality, usability, and effectiveness, indicating strong overall application performance. With regard to the Functionality test, It scored a mean of 4.85 for translating braille text to word accurately (item number 2), and scanning feature processes images quickly and efficiently. Additionally, supporting visually impaired users in their learning activities (item 4 in effectiveness result) obtained a mean of 4.85. These results confirm that the application performs its intended tasks accurately and reliably.

In terms of Usability and effectiveness, the data shows that the users find the application easy to use and navigate, and improves how the users navigate to its surroundings. Item 1 recorded a mean of 4.7 and item 4 garnered a mean score of 4.75, interpreted as Highly Usable. Overall, the results demonstrate that the application is highly functional, effective, and user-friendly with minimal areas of enhancements

### Structure of the Project

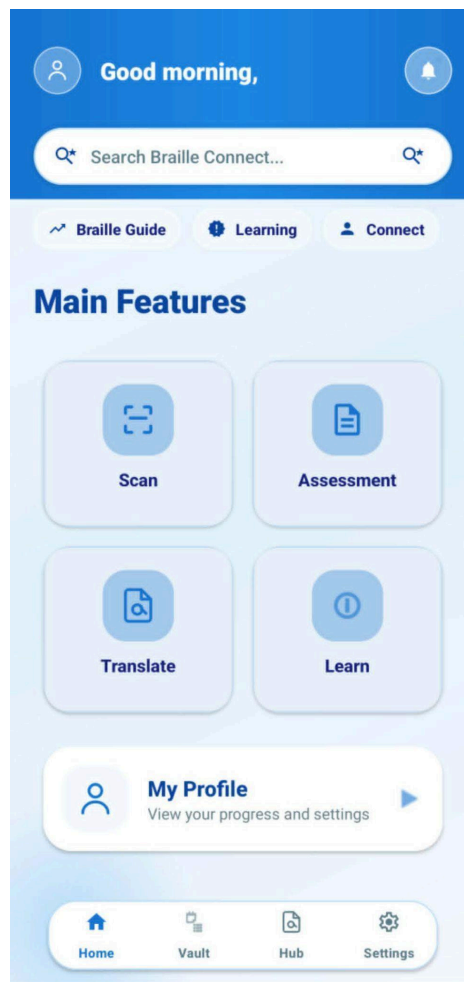


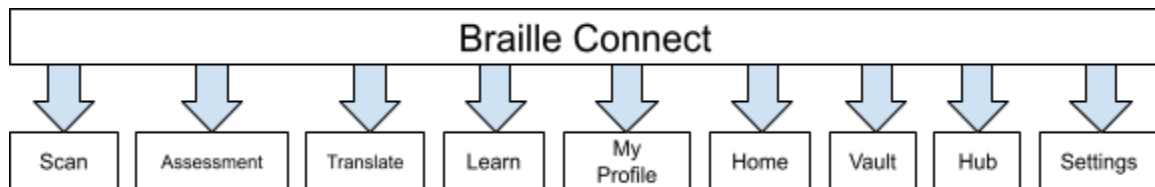
Figure 18. Structure of the Project

| Components        | Functions                                                                                      |
|-------------------|------------------------------------------------------------------------------------------------|
| <b>Scan</b>       | This is used to scan files (e.g. photo, documents)                                             |
| <b>Assessment</b> | This is used for taking assessment of the blind students through uploading the file.           |
| <b>Translate</b>  | This is used for files translation (e.g. uploaded documents, uploaded images, captured photos) |
| <b>Learn</b>      | This is where the user finds the application's summary.                                        |
| <b>My Profile</b> | This is used to display the user's information and progress.                                   |
| <b>Home</b>       | This is where the main functions of the application are located.                               |
| <b>Vault</b>      | This is where the saved files are stored.                                                      |
| <b>Hub</b>        | This is where the additional features are located.                                             |
| <b>Settings</b>   | This is used to enable and disable functions in the application.                               |

*Table 11. Components and Functions*

Below are the features, parts and functions, interrelationship of parts, capabilities, and limitations

### Interrelationship of the Parts



*Figure 19. Interrelationship of the Parts*

The application is divided into various parts, each of which serves an important function during operation.

### Capabilities

The Braille Connect Mobile Application is capable of performing the following operations:

1. Translating documents into audio
2. Allowing users to take assessments through TalkBack Feature
3. Providing gesture and audio-description navigation
4. Saving and storing files within the application
5. Printing test summaries and sending them to the respective teacher

**Limitations**

1. Limited accuracy in color detection and image identification
2. Limited file translation support for only .docx, .pdf, .txt, and .png formats, excluding figures, tables, and graphs
3. Limited access to students only, with no available teacher dashboard
4. Limited functionality to the listed capabilities and inability to perform unlisted functions, such as capturing slide presentations during assessments

**Features/Specification/Technical Effect****Project Features:**

1. Braille-to-word and word-to-Braille translation
2. TalkBack functionality and audio-description support
3. Haptic feedback, gesture controls, and audio beeps for navigation between buttons
4. Voice commands for easier page navigation
5. Assessment mode for uploading and answering assessments within the application
6. Color detection and object identification for recognizing captured colors and objects
7. Assessment summary printing and submission to the respective teacher

## **Operating Procedures**

This section highlights operating instructions, maintenance, safety guidelines, and control measures for the Braille Connect Mobile Application:

### **Operating Instructions:**

These are the steps needed to operate the project:

Step 1. Open the application and wait for 3 seconds for the audio instructions.

Step 2. Navigate the application using gestures and audio descriptions.

Step 3. Swipe Right and Left to move from one tab to another, Swipe Up and Down when moving from one button to another, Tap once to click the desired button.

Step 4. Tap the Scan tab to upload and scan files.

Step 5. Wait for the application to read the uploaded files through TalkBack Feature.

**Maintenance**

To ensure better system performance, the following maintenance procedures should be properly observed:

1. Regularly update the application to the latest version and immediately fix any detected bugs to enhance overall performance.
2. Periodically check the database to ensure that all data are accurate and properly stored.
3. Keep storage devices safe and secure by protecting them from exposure to liquids, excessive heat, and direct sunlight.
4. Create backup storage regularly to prevent data loss or any mishap.
5. Maintain proper control and monitoring of user accounts to ensure the continuous functionality of the application.

**Safety and Control Measure**

This system handles the personal and confidential information of users; therefore, the following safety and control measures are highly recommended to ensure the reliability and security of the application:

1. The application should be used by visually impaired users under the guidance and supervision of parents and teachers.
2. The system should be regularly monitored, updated, and maintained through bug fixes to ensure proper functionality.

## **CHAPTER V**

### **SUMMARY, CONCLUSIONS, AND RECOMMENDATIONS**

#### **Summary of Finding**

The project study titled “Braille Connect: A Mobile Application for Visually Impaired Students” was innovated to address the existing problem of inclusive classroom settings due to a lack of assistive technology that assists teachers in giving classroom instructions and assessments. In the study, it was ascertained that the project was “Highly Functional, Highly Usable, and Highly Effective” as concluded by the teachers, parents, and students within the Matina Pangil Area.

Based on the level of functionality, Items 2 and 4 received the highest ratings with a score of 4.85, interpreted as “highly functional.” Items 1 and 5 ranked second with a score of 4.80, followed by Item 3 with 4.60, also interpreted as “highly functional.”

Moreover, based on the level of usability, Item 4 received the highest rating with a score of 4.75, interpreted as “highly usable.” Item 1 ranked second with 4.70, followed by Item 3 with 4.65, Item 2 with 4.60, and Item 5 with 4.55. All items were interpreted as “highly usable.”

Furthermore, based on the level of effectiveness, Item 2 received the highest rating with a score of 4.85, interpreted as “highly effective.” Items 3 and 4 ranked second with a score of 4.75, followed by Item 1 with 4.65 and Item 5 with 4.55. All items were interpreted as “highly effective.”

Therefore, the least favorable score recorded in functionality is item number 3, which scored 4.6 and was labeled “highly functional,” item 5 for usability, which scored 4.55 and was labeled “highly usable,” and item 5 for effectiveness, which scored 4.55 and was labeled “highly effective.”

## **Conclusions**

Based on the findings of the study, the research project design and development of “Braille Connect: A Mobile Application for Visually Impaired Students” is functional, usable, and effective. The data reveal that the application achieved consistent high mean scores across all three areas (functionality, usability, and effectiveness), indicating that the application can be considered a highly successful and well-developed system. Its strong functionality, usability, and effectiveness suggest that it performs accurately and fulfills its intended goals. Although minor improvements in relation to performance issues may be made to further enhance the user experience, the overall performance confirms that the application is capable of doing its tasks and can be utilized by intended users.

It was shown that the objective of developing the “Braille Connect: A Mobile Application for Visually Impaired Students” was accomplished by this innovation.

## **Recommendations**

Based on the analysis the following suggestions are recommended:

1. Add a looping carousel navigation concept.
2. Ensure availability on IOS.
3. Provide additional gesture functions different from others to avoid overlapping functionality.
4. Improve the accuracy of the Color Detector feature.
5. Add a feature where users can take a photo of the quiz on the board, and the blind students can take the quiz using the application.

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## **APPENDICES**

## Appendix A

## Adviser Acceptance Form



**CITY COLLEGE OF DAVAO**

### Bachelor of Technical-Vocational Teacher Education

📍 Km. 10, Catalunan Pequeno, Davao City 📞 +63 (82) 241 7310 📧 admin@cd.edu.ph 🌐 facebook.com/ChyCollegeDavaoCity

Date:

**DR. IDA G. Tudy**

**Program Head, Bachelor of Technical Vocational Teacher Education**

**Dear Dr. Tudy:**

The Bachelor of Technical-Vocational Teacher Education (BTVTEd) student-candidates, majoring in Computer Programming/HVACR Technology, are currently undertaking the Thesis/Project Study requirement of the program. We have identified the topic entitled, "Braille.Connect: An Automated Braille Device with Software", which we intend to pursue as our thesis/project study for the First Semester of the Academic Year 2025-2026.

In line with this, we respectfully request the services of Mr./Ms. Wendell Galgo as our Project Study Adviser to guide us in the successful completion of this academic endeavor.

I shall highly appreciate your favorable response.

Sincerely,

Neil Francis T. Arnaiz

Zaldy ~~A. Dagohoy~~

(Signature over printed name)

Conforme: Wendell Galgo

Thesis Adviser

*"Dedicated to Excellence, Committed to Service"*



## Appendix B

### Permission Letter




# CITY COLLEGE OF DAVAO

Bachelor of Technical-Vocational Teacher Education

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**07 APRIL 2026**

**DR. WENEFREDO E. CAGAPE**  
 College President  
 City College of Davao

Dear Sir,

We are 3<sup>rd</sup>-Year students from **City College of Davao**, currently enrolled in the subject **Technology Research**. As part of our requirements for this course, we are developing a research project entitled "**Braille Connect: A Software Application for Visually Impaired Students**".

We are writing to respectfully request your permission to gather data at **Davao School for the Blinds, Barangay Bago Aplaya, Davao City**. This process is a crucial step in ensuring that our innovation's features is effective and reliable.

The objective of our study is to:

- **Test the functionality of the application in terms of:**
  - *Scanning Images*
  - *Talkback Feature of the application*
  - *Storing uploaded documents (e.g. docs, pdf, texts)*
- **Test the usability of the application in terms of:**
  - *Application Controls*
  - *Audio Output*
  - *Application Interface*
- **Test the effectiveness of the application in terms of:**
  - *Objects Identification*

Our questionnaire aims to conduct a survey to at least no more than **20** from **Davao School for the Blinds, Bago Aplaya, Davao City**. We assure you that the data gathered will be handled with the strictest confidentiality and will be used solely for academic purposes within our Technology Research Innovation subject.

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## Appendix B.1

### Permission Letter




# CITY COLLEGE OF DAVAO

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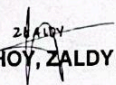
Your support would greatly assist us in refining our research and contributing to the success of our innovation. Attached is a copy of our questionnaire and Informed Consent Form (ICF) for your reference and review.

Thank you for your time and for supporting the student-researchers of the City College of Davao. We look forward to your favorable response.

Sincerely,

  
**ARNAIZ, NEIL FRANCIS T.**  
*Researcher*

  
**DAGOHOY, ZALDY A.**  
*Researcher*

Noted by:

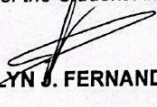
  
**MR. WENDELL F. GALGO**  
*Research Adviser*

  
**DR. IDA G. TUDY**  
*Program Head, BTVTEd*

Recommending Approval:

  
**DR. ISMAEL W. BAOG**  
*Head, Office of the Student Affairs*

  
**DR. EDROSLYN J. FERNANDEZ**  
*Vice President for Academic Affairs*

Approved by:

  
**DR. WENEFREDO E. CAGAPE**  
*College President*

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## Appendix C

### Informed Consent Letter




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**Ethics Informed Consent**

**Form (ICF) Informed Consent Form for**

**Student/Instructor/Parent**

**Name of the Researcher(s) Neil Francis T. Arnalz and Zaldy A. Dagohoy**

**Institution: CITY COLLEGE OF DAVAO**

**INTRODUCTION**

You are invited to participate in a research study entitled "Braille Connect: A Software Application" conducted at the City College of Davao. You are selected because you are a visually impaired student, parent/guardian, and teacher involved in the learning process of students with visual impairments, and your insights are vital for this study.

Your participation is entirely voluntary. Please take your time to read this form carefully and feel free to ask questions. If you agree to participate, please sign below. You will receive a copy of this form.

**PURPOSE OF THE STUDY**

The purpose of this study is to Design and Develop Braille Connect: A Software Application, convert braille texts, documents, and captured photos through Audio , assess the accuracy and functionality of the translation feature from Braille texts, documents, and captured photos, and evaluate the effectiveness of the application's ability to translate and upload document files such as Docs, PDF, etc. Furthermore, it aims to improve accessibility of learning materials for Visually Impaired Students.

**STUDY PROCEDURES**

You will be asked to use the Braille Connect Application during a testing session conducted in a classroom or certain controlled environment.

---

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## Appendix C.1

### Informed Consent Letter




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---

**During the session, you will:**

- Use the application to capture photos and/or upload document files and translate it through speech.
- Explore and navigate the features of the application such as Braille text-to-speech translation, file uploading, and captured photo description.
- Participate in an actual classroom discussion using the application.
- Provide feedback through a short survey or a brief interview

**After completing the survey:**

- **If onsite:** You will submit the completed form directly to the researcher.
- **If online:** You will submit your responses through the secure link provided or send them to the researcher using the specified method.
- The entire session will take approximately **15–20 minutes**, depending on your role and assigned tasks.

**Before Testing**

- You will be given an **Informed Consent Form (ICF)** that:
  - Explains the purpose of this phase of the study.
  - Confirms that your participation is voluntary.
  - States your right to withdraw from the study at any time without penalty.
  - Assures you that your grades or job performance evaluations will not be affected by your participation.

**Testing Setup**

- **Onsite:** The activity will take place in a classroom, where the researcher will be present to provide guidance and answer questions.
- **Online (if necessary):** The activity will be conducted through secure access to the tool, with instructions provided beforehand.

**Data Handling**

- Your responses will be anonymized using participant codes.
- All collected data will be stored securely for **two (2) years**.
- After this period, all data will be permanently deleted from all systems and backups.
- Only members of the research team will have access to the data.

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## Appendix C.2

### Informed Consent Letter




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---

**Your Comfort**

- You may experience minor discomforts such as screen fatigue or eye strains for non-visually impaired participants.
- You are free to take breaks, skip any activity, or withdraw from the study at any time.

**POTENTIAL RISKS AND DISCOMFORTS**

- This study does not involve sensitive or invasive topics. However, you may experience minor discomforts such as mental fatigue, reduced concentration, or slight inconvenience due to the time needed to complete the activities (approximately **15–20 minutes**, depending on your role as a student, teacher, or administrator).
- You may take short breaks, skip any question, or withdraw from the study at any time if you feel uncomfortable.
- The software being tested will not influence your academic standing, grades, or employment status.
- Participation in this study is entirely voluntary, and your privacy and confidentiality will be respected at all times.
- The safety, comfort, and well-being of participants will remain the top priority throughout the study.

**DATA PRIVACY AND CONFIDENTIALITY**

This study ensures the protection of participants' privacy and confidentiality in accordance with the **Data Privacy Act of 2012 (RA 10173)**. Participants' identities will remain anonymous through the use of coded identifiers instead of real names. All collected data will be securely stored in password-protected and encrypted files accessible only to the researcher and will be retained for two (2) years before being permanently deleted. Any research results that are published or presented will include only aggregated and anonymized data, ensuring that no personally identifiable information is disclosed.

**VOLUNTARINESS OF PARTICIPATION AND RIGHT TO WITHDRAW FROM THE RESEARCH**

Your participation in this study is entirely voluntary. Choosing not to participate will not result in any penalty or loss of benefits to which you are otherwise entitled. You may also withdraw your consent and discontinue your participation at any time without any penalty or negative consequences. By participating in this research, you are not waiving any legal claims, rights, or remedies.

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## Appendix C.3

### Informed Consent Letter




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**RESEARCH PARTICIPANT'S CONSENT**

I have read and understood the information provided above. I have been given the opportunity to ask questions, and all my questions have been answered satisfactorily. I voluntarily agree to participate in this study. I have also received a copy of this consent form. I understand that I may withdraw my consent and discontinue my participation at any time without any penalty.

Signature of Participant: \_\_\_\_\_  
 Printed Name of Participant: \_\_\_\_\_  
 Date Signed: \_\_\_\_\_

To be accomplished by the Researcher Obtaining Consent:

I have explained the research to the participant and answered all of his/her questions. I believe that he/she understands the information described in this document and freely consents to participate.

| Name of Person Obtaining Consent | Date Signed |
|----------------------------------|-------------|
|                                  |             |

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## Appendix D

### Endorsement for Title Defense



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October 29, 2025  
Date

IDA G. TUDY, PhD  
Program Head, BTVTED

*Received  
10/29/25*

Maam:

Respectfully informing your good Office that ARWALIZ, GROUP, a 3<sup>rd</sup> year student of the BTVTED major in CP/HVACRT program of this College is ready to:

( ☒ ) defend thesis proposal

( ☐ ) undergo thesis final oral defense

Entitled: Braille Connect: An automated Braille Device With Software Application

on: November 11, 2025

1-2 PM

BOT Room

Date

Time

Venue

Sincerely,

WENDEL  
Thesis Adviser

Conforme:

Panel Members

Signature

Date

ORA CANTERA /IMCY KATE

URGIND, EMEILIA Z.

DOMA R. SAAZAR

10/29/25

10/29/2025

10/29/2025

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## Appendix E

### Proposal Manuscript Approval



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### PROPOSAL MANUSCRIPT APPROVAL

**Name of Student:** Amaiz, Neil Francis T. & Dagohoy, Zaldy A.  
**Program:** Bachelor of Technical-Vocational Teacher Education  
**Thesis Title:** Braille Connect: A Software Application for Visually Impaired Students

**Adviser:** Wendell F. Galgo

| Panel Member's Name            | Comments/ Suggestions                                                                                                                                                                                                                                                                            | Action Taken (with Page Number)                                                                                                                                           | Panel Member's Signature |
|--------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------|
| Mr. Romel P. Salazar           | Focus only on one specific innovation. Whether focusing on just the device or the Software application.                                                                                                                                                                                          | (a) Inform the panelist for the letter of change<br><br>(b) Revise the paper focusing on one innovation<br><br>(c) Revise the thesis title appropriate to the new concept |                          |
| Ms. Honey Kate A. Dela Calzada | (a) Focus first on synchronization of the software to the braille device, but do not limit.<br><br>(b) Include the description of the new process flow<br><br>(c) Specify the research tool to be used<br><br>(d) limit the number of the participants to 5 students, 5 parents, and 5 teachers. | (a) Add and specify the objective of our study; pg 4.<br><br>(b) Add the description of the new process flow; pg 29.<br><br>(c) Revised the Research Instruments; pg 36.  |                          |
| Ms. Emeliza Z. Urgino          | Determine and specify how to administer the testing process                                                                                                                                                                                                                                      | Determine the research tool will be used; pg 36.                                                                                                                          |                          |

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## Appendix F

### Request Letter for Change




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**07 APRIL 2026**

**REQUEST LETTER FOR CHANGE**

Good day Ma'am/Sir,

I am writing to formally request approval to revise our research entitled ***"Braille Connect: An Automated Braille Device with Software Application"***.

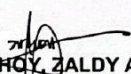
After further evaluation and guidance, we found it necessary to apply certain changes to improve the clarity and quality of the study. These revisions are intended to strengthen the research and ensure that it meets academic standards.

We assure you that the proposed changes will not affect the validity of the study but will help enhance its overall outcome. We are prepared to follow any instructions or requirements regarding this request.

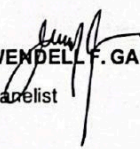
Thank you very much for your time and consideration. We hope for your kind approval.

Respectfully yours,

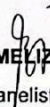
  
**ARNAIZ, NEIL FRANCIS T.**  
 Researcher

  
**DAGOHUY, ZALDY A.**  
 Researcher

Noted by:

  
**WENDELL F. GALGO**  
 Panelist

  
**HONEY KATE DELA CALZADA**  
 Panelist

  
**EMELIZA Z. URGINO**  
 Panelist

Approved by:

  
**DR. ROMEL P. SALAZAR**  
 Research Instructor

---

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## Appendix G

### Survey Questionnaire



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### SURVEY QUESTIONNAIRE

#### Braille Connect: A Mobile Application for Visually Impaired Students

Name (Optional): \_\_\_\_\_

Respondent Group:

- ☐ Faculty  
☐ Parent  
☐ Student

Please rate the attached questionnaire by checking the appropriate box, guided by the following codes:

#### The Level of Functionality of the Braille Connect Scanning Application

5 – Highly Functional 4 – Functional 3 – Moderately Functional 2 – Slightly Functional 1 – Non Functional

|   | Contents                                                                                                                                                                                  | Rating |   |   |   |   |
|---|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|---|---|---|---|
|   |                                                                                                                                                                                           | 5      | 4 | 3 | 2 | 1 |
| 1 | When scanning the surroundings, the application captures images clearly and accurately.<br><i>(Kung mag-scan sa palibot, ang aplikasyon makakuha og mga hulagway nga klaro ug tukma.)</i> |        |   |   |   |   |
| 2 | The system accurately translate word-to-braille<br><i>(Ang sistema tukmang naghubad sa mga pulong ngadto sa Braille.)</i>                                                                 |        |   |   |   |   |
| 3 | The application generates clear and easy-to-understand image descriptions.<br><i>(Ang aplikasyon nagmugna og klaro ug sayon sabton nga mga deskripsyon sa hulagway.)</i>                  |        |   |   |   |   |
| 4 | The scanning feature processes images quickly and efficiently.<br><i>(Ang pag-scan nga feature makaproseso sa mga hulagway dali ug episyente.)</i>                                        |        |   |   |   |   |

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## Appendix G.1

### Survey Questionnaire



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|            |                                                                                                                                                                               |  |  |  |  |  |
|------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--|--|--|--|
| 5          | <b>The TalkBack feature produces clear and audible voice descriptions.</b><br><br><i>(Ang TalkBack feature mopagawas og klaro ug madungog nga mga deskripsyon sa tingog.)</i> |  |  |  |  |  |
|            |                                                                                                                                                                               |  |  |  |  |  |
| Mean score |                                                                                                                                                                               |  |  |  |  |  |

What functionality do you think is important ?

*(Unsa nga functionality sa imong tan-aw ang importante?)*

---



---

#### The Level of Usability of the Braille Connect Scanning Application

5 – Highly Usable 4 – Usable 3 – Moderately Usable 2 – Slightly Usable

1 – Non Usable

| Contents                                                                                                                                                                          | Rating |   |   |   |   |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|---|---|---|---|
|                                                                                                                                                                                   | 5      | 4 | 3 | 2 | 1 |
| <b>1 The application interface is accessible for visually impaired users.</b><br><br><i>(Ang interface sa aplikasyon dali ma-access para sa mga adunay problema sa panan-aw.)</i> |        |   |   |   |   |
| <b>2 The controls of the application are easy to understand and operate.</b><br><br><i>(Ang mga kontrol sa aplikasyon sayon sabton ug sayon gamiton.)</i>                         |        |   |   |   |   |
| <b>3 The scanning feature can be used without difficulty by blind students.</b><br><br><i>(Ang scanning feature mahimong magamit sa walay kalisod sa mga estudyanteng buta.)</i>  |        |   |   |   |   |

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## Appendix G.2

### Survey Questionnaire



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|                                                                                                                                                                                                                                         |  |  |  |  |  |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--|--|--|--|
| 4 The audio output of the application is clear and easy to follow.<br><br>(The audio output of the application is clear and easy to follow.)                                                                                            |  |  |  |  |  |
| 5 The application functions without lag, crashes, or performance issues during extended and continuous use.<br><br>(Ang aplikasyon molihok nga walay lag, pag-crash, o problema sa performance bisan sa dugay ug padayon nga paggamit.) |  |  |  |  |  |
| Mean score                                                                                                                                                                                                                              |  |  |  |  |  |

Is the mobile application useful for both teachers and students?

(Mapuslanon ba ang mobile application para sa mga magtutudlo ug mga estudyante?)

---



---



---

#### The Level of Effectiveness of the Braille Connect Scanning Application

5 – Highly Effective 4 – Effective 3 – Moderately Effective 2 – Slightly Effective

1 – Non Effective

| Contents                                                                                                                                                                                                 | Rating |   |   |   |   |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|---|---|---|---|
|                                                                                                                                                                                                          | 5      | 4 | 3 | 2 | 1 |
| 1 The application helps visually impaired users identify objects in their surroundings.<br><br>(Ang aplikasyon makatabang sa mga adunay problema sa panan-aw sa pag-ila sa mga butang sa ilang palibot.) |        |   |   |   |   |

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## Appendix G.3

### Survey Questionnaire



## CITY COLLEGE OF DAVAO

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|            |                                                                                                                                                                                                                  |  |  |  |  |  |
|------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--|--|--|--|
| 2          | <b>The application supports visually impaired students in their learning activities.</b><br><br>(Ang aplikasyon nagasuporta sa mga estudyanteng adunay problema sa panan-aw sa ilang mga kalihokan sa pagtuon.)  |  |  |  |  |  |
| 3          | <b>The system assists teachers and parents in guiding visually impaired students.</b><br><br>(Ang sistema makatabang sa mga magtutudlo ug ginikanan sa paggiya sa mga estudyanteng adunay problema sa panan-aw.) |  |  |  |  |  |
| 4          | <b>The application improves the user's awareness of their environment.</b><br><br>(Ang aplikasyon nagpauswag sa kahibalo sa user bahin sa ilang palibot.)                                                        |  |  |  |  |  |
| 5          | <b>The application contributes to the independence of visually impaired users.</b><br><br>(Ang aplikasyon makatabang sa pagpalig-on sa kagawasan sa mga user nga adunay problema sa panan-aw.)                   |  |  |  |  |  |
| Mean score |                                                                                                                                                                                                                  |  |  |  |  |  |

Does the mobile application effectively assist both teachers and students?

(Epektibo ba ang mobile application sa pagtabang sa mga magtutudlo ug mga estudyante?)

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## Appendix H

### Validation of Survey Questionnaire



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### QUESTIONNAIRE VALIDATION FORM

Name of evaluator: WENDEL F. CALAO

Highest degree: MIT (CAR)

Position: IT-IN-CHARGE, CCD

Please rate the attached questionnaire by checking the appropriate box, guided by the following codes:

5 – Excellent 4 – Very good 3 – Good 2 – Fair 1 – Poor

| Contents                                                                                                                                                                                                                              | Rating                              |   |   |   |   |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------|---|---|---|---|
|                                                                                                                                                                                                                                       | 5                                   | 4 | 3 | 2 | 1 |
| 1 The instructions and test items are clearly stated. The vocabulary, language structure, and conceptual difficulty of the questions are appropriate for the participants' level, making the directions and items easy to understand. | <input checked="" type="checkbox"/> |   |   |   |   |
| 2 The items are presented clearly and arranged in a logical and coherent order.                                                                                                                                                       | <input checked="" type="checkbox"/> |   |   |   |   |
| 3 The items suitably reflect the content of the research. The questions are constructed to effectively measure the intended conditions, knowledge, perceptions, and attitudes.                                                        | <input checked="" type="checkbox"/> |   |   |   |   |
| 4 The items sufficiently cover the scope of the research. Each category contains an adequate number of questions to represent all the necessary aspects required for the study.                                                       | <input checked="" type="checkbox"/> |   |   |   |   |
| 5 The instrument, as a whole, effectively achieves the objectives for which it was developed.                                                                                                                                         | <input checked="" type="checkbox"/> |   |   |   |   |
| 6 Each item is designed to elicit a single, specific response or measure only one behavior, and no part of the questionnaire indicates researcher bias.                                                                               | <input checked="" type="checkbox"/> |   |   |   |   |
| 7 The adopted scale and rating system are appropriate for the items being evaluated.                                                                                                                                                  | <input checked="" type="checkbox"/> |   |   |   |   |
| Mean score                                                                                                                                                                                                                            |                                     |   |   |   |   |

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## Appendix H.1

### Validation of Survey Questionnaire



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### QUESTIONNAIRE VALIDATION FORM

Name of evaluator: SALAZAR ROMEL P

Highest degree: PhD

Position: IT

Please rate the attached questionnaire by checking the appropriate box, guided by the following codes:

5 – Excellent 4 – Very good 3 – Good 2 – Fair 1 – Poor

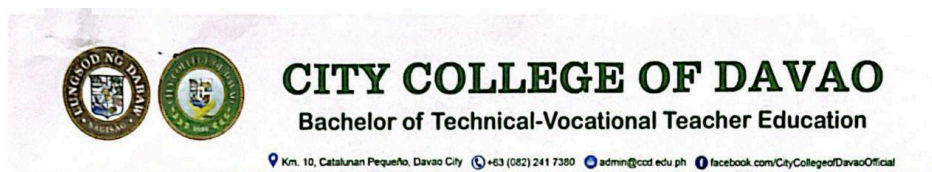
| Contents                                                                                                                                                                                                                              | Rating |   |   |   |   |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|---|---|---|---|
|                                                                                                                                                                                                                                       | 5      | 4 | 3 | 2 | 1 |
| 1 The instructions and test items are clearly stated. The vocabulary, language structure, and conceptual difficulty of the questions are appropriate for the participants' level, making the directions and items easy to understand. |        | / |   |   |   |
| 2 The items are presented clearly and arranged in a logical and coherent order.                                                                                                                                                       | /      |   |   |   |   |
| 3 The items suitably reflect the content of the research. The questions are constructed to effectively measure the intended conditions, knowledge, perceptions, and attitudes.                                                        | /      |   |   |   |   |
| 4 The items sufficiently cover the scope of the research. Each category contains an adequate number of questions to represent all the necessary aspects required for the study.                                                       | /      |   |   |   |   |
| 5 The instrument, as a whole, effectively achieves the objectives for which it was developed.                                                                                                                                         |        | / |   |   |   |
| 6 Each item is designed to elicit a single, specific response or measure only one behavior, and no part of the questionnaire indicates researcher bias.                                                                               | /      |   |   |   |   |
| 7 The adopted scale and rating system are appropriate for the items being evaluated.                                                                                                                                                  | /      |   |   |   |   |
| Mean score                                                                                                                                                                                                                            |        |   |   |   |   |

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## Appendix H.2

### Validation of Survey Questionnaire



#### QUESTIONNAIRE VALIDATION FORM

Name of evaluator: EMELITA Z. URGIND

Highest degree: MTE

Position: ASSISTANT PROFESSOR I

Please rate the attached questionnaire by checking the appropriate box, guided by the following codes:

5 – Excellent    4 – Very good    3 – Good    2 – Fair    1 – Poor

| Contents                                                                                                                                                                                                                              | Rating |   |   |   |   |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|---|---|---|---|
|                                                                                                                                                                                                                                       | 5      | 4 | 3 | 2 | 1 |
| 1 The instructions and test items are clearly stated. The vocabulary, language structure, and conceptual difficulty of the questions are appropriate for the participants' level, making the directions and items easy to understand. |        | / |   |   |   |
| 2 The items are presented clearly and arranged in a logical and coherent order.                                                                                                                                                       |        |   | / |   |   |
| 3 The items suitably reflect the content of the research. The questions are constructed to effectively measure the intended conditions, knowledge, perceptions, and attitudes.                                                        |        |   | / |   |   |
| 4 The items sufficiently cover the scope of the research. Each category contains an adequate number of questions to represent all the necessary aspects required for the study.                                                       |        |   | / |   |   |
| 5 The instrument, as a whole, effectively achieves the objectives for which it was developed.                                                                                                                                         |        |   | / |   |   |
| 6 Each item is designed to elicit a single, specific response or measure only one behavior, and no part of the questionnaire indicates researcher bias.                                                                               | /      |   |   |   |   |
| 7 The adopted scale and rating system are appropriate for the items being evaluated.                                                                                                                                                  | /      |   |   |   |   |
| Mean score                                                                                                                                                                                                                            |        |   |   |   |   |

*Pls go back to the objectives of your study. I believe the objectives only included functionality.*

## Appendix H.3

### Validation of Survey Questionnaire




# CITY COLLEGE OF DAVAO

## Bachelor of Technical-Vocational Teacher Education

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### QUESTIONNAIRE VALIDATION FORM

Name of evaluator: DELA CRISTINA, HK

Highest degree: \_\_\_\_\_

Position: INSTRUCTOR I

Please rate the attached questionnaire by checking the appropriate box, guided by the following codes:

5 – Excellent    4 – Very good    3 – Good    2 – Fair    1 – Poor

| Contents                                                                                                                                                                                                                              | Rating |   |   |   |   |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|---|---|---|---|
|                                                                                                                                                                                                                                       | 5      | 4 | 3 | 2 | 1 |
| 1 The instructions and test items are clearly stated. The vocabulary, language structure, and conceptual difficulty of the questions are appropriate for the participants' level, making the directions and items easy to understand. | /      |   |   |   |   |
| 2 The items are presented clearly and arranged in a logical and coherent order.                                                                                                                                                       | /      |   |   |   |   |
| 3 The items suitably reflect the content of the research. The questions are constructed to effectively measure the intended conditions, knowledge, perceptions, and attitudes.                                                        |        | / |   |   |   |
| 4 The items sufficiently cover the scope of the research. Each category contains an adequate number of questions to represent all the necessary aspects required for the study.                                                       |        | / |   |   |   |
| 5 The instrument, as a whole, effectively achieves the objectives for which it was developed.                                                                                                                                         | /      |   |   |   |   |
| 6 Each item is designed to elicit a single, specific response or measure only one behavior, and no part of the questionnaire indicates researcher bias.                                                                               |        | / |   |   |   |
| 7 The adopted scale and rating system are appropriate for the items being evaluated.                                                                                                                                                  | /      |   |   |   |   |
| Mean score                                                                                                                                                                                                                            |        |   |   |   |   |

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## Appendix I

### Certificate of Statistician



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### CERTIFICATE OF STATISTICIAN

This is to certify that I have reviewed, checked, and analyzed the statistical data and procedures used in the research study entitled:

**“Braille Connect: A Mobile Application for Visually Impaired Students”**

conducted by the students from the Bachelor of Technical Vocational Teacher Education Major in Computer Programming of City College of Davao.

I hereby affirm that the statistical tools, methods, and interpretations applied in this study are appropriate, accurate, and in accordance with standard statistical practices. The data analysis has been carefully examined and validated to ensure the reliability and correctness of the results presented in this research.

This certification is issued upon the request of the researchers for whatever legal and academic purposes it may serve.

Certified by:

**MR. ALBERT GRIAR**  
Statistician

Date: May 5, 2026

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## Appendix J

### Certificate of Grammarian

#### CERTIFICATE OF LANGUAGE EDITING

This is to certify that the undersigned has checked and reviewed the thesis titled:

**BRAILLE CONNECT: A MOBILE APPLICATION FOR VISUALLY IMPAIRED  
STUDENTS**

prepared by **NEIL FRANCIS T. ARNAIZ** and **ZALDY A. DAGOHAY**, and has found it satisfactory according to the set of structural rules that govern the composition of sentences, phrases, and words in the English Language.

  
**KAREN MAY C. DOTOLLO, PhD**  
Grammarian/Language Editor  
PRC License Number: 1285934

May 26, 2026

## Appendix K

### AI Report



Page 2 of 51 - AI Writing Overview

Submission ID tm:oid::31188:136657338

#### \*% detected as AI

AI detection includes the possibility of false positives. Although some text in this submission is likely AI generated, scores below the 20% threshold are not surfaced because they have a higher likelihood of false positives.

Caution: Review required.

It is essential to understand the limitations of AI detection before making decisions about a student's work. We encourage you to learn more about Turnitin's AI detection capabilities before using the tool.

#### Disclaimer

Our AI writing assessment is designed to help educators identify text that might be prepared by a generative AI tool. Our AI writing assessment may not always be accurate (i.e., our AI models may produce either false positive results or false negative results), so it should not be used as the sole basis for adverse actions against a student. It takes further scrutiny and human judgment in conjunction with an organization's application of its specific academic policies to determine whether any academic misconduct has occurred.

#### Frequently Asked Questions

##### How should I interpret Turnitin's AI writing percentage and false positives?

The percentage shown in the AI writing report is the amount of qualifying text within the submission that Turnitin's AI writing detection model determines was either likely AI-generated text from a large-language model or likely AI-generated text that was likely revised using an AI paraphrase tool or word spinner.

False positives (incorrectly flagging human-written text as AI-generated) are a possibility in AI models.

AI detection scores under 20%, which we do not surface in new reports, have a higher likelihood of false positives. To reduce the likelihood of misinterpretation, no score or highlights are attributed and are indicated with an asterisk in the report (\*%).

The AI writing percentage should not be the sole basis to determine whether misconduct has occurred. The reviewer/instructor should use the percentage as a means to start a formative conversation with their student and/or use it to examine the submitted assignment in accordance with their school's policies.

##### What does 'qualifying text' mean?

Our model only processes qualifying text in the form of long-form writing. Long-form writing means individual sentences contained in paragraphs that make up a longer piece of written work, such as an essay, a dissertation, or an article, etc. Qualifying text that has been determined to be likely AI-generated will be highlighted in cyan in the submission, and likely AI-generated and then likely AI-paraphrased will be highlighted purple.

Non-qualifying text, such as bullet points, annotated bibliographies, etc., will not be processed and can create disparity between the submission highlights and the percentage shown.



Page 2 of 51 - AI Writing Overview

Submission ID tm:oid::31188:136657338

# Appendix L

## Similarity Report



Page 2 of 74 - Integrity Overview

Submission ID: trnoid::31188139435982

### 10% Overall Similarity

The combined total of all matches, including overlapping sources, for each database.

#### Filtered from the Report

- Bibliography
- Quoted Text
- Cited Text
- Small Matches (less than 8 words)

#### Exclusions

- 13 Excluded Matches

#### Match Groups

- 71 Not Cited or Quoted 10%

Matches with neither in-text citation nor quotation marks
- 0 Missing Quotations 0%

Matches that are still very similar to source material
- 0 Missing Citation 0%

Matches that have quotation marks, but no in-text citation
- 0 Cited and Quoted 0%

Matches with in-text citation present, but no quotation marks

#### Top Sources

- 3% Internet sources
- 1% Publications
- 8% Submitted works (Student Papers)

#### Integrity Flags

##### 1 Integrity Flag for Review

**Hidden Text**  
15 suspect characters on 12 pages  
Text is altered to blend into the white background of the document.

Our system's algorithms look deeply at a document for any inconsistencies that would set it apart from a normal submission. If we notice something strange, we flag it for you to review.

A Flag is not necessarily an indicator of a problem. However, we'd recommend you focus your attention there for further review.



Page 2 of 74 - Integrity Overview

Submission ID: trnoid::31188139435982

## Appendix M

### Documentation



## Appendix N

### Certificate of Originality



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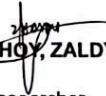
#### CERTIFICATE OF ORIGINALITY

We hereby certify that this thesis is the result of our own original work and has not been submitted, either in whole or in part, for the award of any degree, diploma, or other academic recognition in any institution of higher learning. To the best of our knowledge, this manuscript does not contain any material previously written or published by another author, except where proper acknowledgment and citation have been duly indicated in the text.

We further affirm that the intellectual content, analysis, findings, and conclusions presented in this study are products of our own effort and understanding, although assistance may have been received in terms of grammar, formatting, presentation, and language enhancement.

  
ARNAIZ, NEIL FRANCIS T.

Researcher

  
DAGOHAY, ZALDY A.

Researcher

  
WENDELL F. GALGO

Researcher Adviser

May 26, 2026

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## **CURRICULUM VITAE**

## Curriculum Vitae



### NEIL FRANCIS ARNAIZ

A passionate and dedicated student leader with strong communication, leadership, and organizational skills. Experienced in student governance, event coordination, and community-driven initiatives, with a commitment to promoting inclusivity, teamwork, and innovation. Skilled in collaborating with diverse groups and managing responsibilities in fast-paced environments. Eager to continuously learn, grow professionally, and contribute meaningful ideas that create a positive impact within the community.

#### CONTACT

##### Phone

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##### Email

neilfrancistravilla.arnaiz@ccd.edu.ph

##### Address

Km. 7, Matina Pangl, Davao City

#### EDUCATION

2023 - 2027

City College of Davao

- BTVTED Major in Computer Programming

2018 - 2020

Daniel R. Aguinaldo National High School

- Science, Technology, Engineering and Mathematics

#### KEY SKILLS

- Design Tools
- Website and Application Development
- UX/UI Design
- Proficient in MS Office
- Proficient in Audio-Visual Editing Software
- Responsive Web Design
- Team Collaboration

#### WORK EXPERIENCE

- City College of Davao** 2025- PRESENT  
**Supreme Student Government - President**
  - Leads the student government and represents the student body.
  - Oversees operations, plans activities, and supervises officers.
  - Represents student concerns and promotes student welfare.
- City College of Davao** 2023 - 2025  
**Supreme Student Government - Assistant Treasurer**
  - Assisted in managing funds and maintaining financial records.
  - Supported budgeting, collections, and disbursements.
  - Helped ensure transparency and accuracy in financial transactions.

#### ACHIEVEMENTS

- School Ambassador** 2024  
 advocate for inclusive education, a learning environment where every student is accepted, respected, and given equal opportunities regardless of their abilities or background. I believe no learner should feel excluded or unheard. Every student deserves a safe space where they can confidently express themselves and reach their full potential.
- Face of CCD** 2023  
 To promote the idea that true beauty comes from the heart and not just from physical appearance, I aim to inspire others through kindness, confidence, and authenticity. I believe that beauty is reflected in how we treat people, uplift others, and create a positive impact within our community. True confidence comes from embracing who we are and encouraging others to do the same.

#### REFERENCE

**Romel P. Salazar**  
 Research Coordinator  
 Phone: +63 (082) 308 9147  
 Email : r.salazar@ccd.edu.ph

**Ida G. Tudy**  
 Program Head  
 Phone: +63 (082) 308 9147  
 Email : itudy@ccd.edu.ph

## Curriculum Vitae



## ZALDY DAGOHEY

Dedicated and motivated student leader with strong experience in leadership, communication, and campus involvement. Served as a Supreme Student Government officer with a passion for organizing events, promoting student welfare, and leading collaborative initiatives. Skilled in problem-solving, teamwork, and project development, with interests in technology, innovation, and community-driven programs. Committed to continuous learning and using creativity and leadership to contribute positively to both academic and professional environments.

### CONTACT

#### Phone

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Bangkas Heights 7-A, Toril Davao City

### EDUCATION

2023 - 2027

City College of Davao

- BTVTED Major in Computer Programming

2021 - 2022

Catalunan Pequeno National High School

- Humanities and Social Sciences

### KEY SKILLS

- Design Tools
- Development
- UX/UI Design
- Wireframing
- Prototyping
- Responsive Web Design
- SEO Best Practices
- Problem-Solving
- Team Collaboration
- Project Management

### WORK EXPERIENCE

City College of Davao 2025- PRESENT  
Supreme Student Government - Vice President

- Assist in leading and managing Supreme Student Government operations and activities.
- Represent students and communicate their concerns to the administration.
- Help organize and supervise school events, programs, and student initiatives.

City College of Davao 2024 - PRESENT  
SIBYA School Publication - Editorial Cartoonist

- Creates visual editorial cartoons that express opinions on current school issues and events.
- Translates complex topics into engaging, thought-provoking illustrations for publication.
- Collaborates with the editorial team to align cartoons with the publication's message and tone.

### ACHIEVEMENTS

Mr Fitness 2024

To inspire the youth to embrace fitness, discipline, and mental wellness as foundations for personal growth and success. I believe that sports and healthy living build confidence, resilience, and unity, empowering students to become responsible individuals who lead by example and motivate others to strive for a better and healthier future.

Mr CCD Intramurals - Second Runner Up 2024

to promote equality and inclusivity by encouraging every student to feel valued, respected, and empowered regardless of gender, background, appearance, or abilities. I believe that true strength is shown not by superiority over others, but by the ability to uplift, accept, and stand alongside everyone with fairness and compassion. Through sports and leadership, we can build a campus community where everyone has an equal opportunity to grow, participate, and succeed.

### REFERENCE

Romel P. Salazar  
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